16.

Problems Based on Age

Type - 6

1. The difference of age of father and son is 24 years. Two years before the age of father was as the twice of the present age of the son what is the present age of the father?

(a) 46 (b) 42
(c) 44 (d) 38

RRB JE - 24/05/2019 (Shift-II)

Ans : (a) Let the present age of father is = x years
and present age of son = y years
difference between the age of father and son = 24 years

$x - y = 24$ ----(i)

According to question,

$x - 2 = 2y$

$x - 2y = 2$ ----(ii)

from eq^n (i) and (ii)–

$x - y = 24$
$\underline{x - 2y = 2}$
$y = 22$ years

Putting the value of y in eq^n (i)

$x - y = 24$
$x - 22 = 24$
$x = 46$

Hence, Present age of father = 46 years

2. The ratio of age of mother and daughter is 9:2. At the time of daughters birth, mothers age was 28 years. What is the age of daughter?

(a) 12 Years (b) 8 Years
(c) 4 Years (d) 6 Years

RRB JE - 27/05/2019 (Shift-II)

Ans : (b) Let the age of daughter = 2x years
Age of mother = 9x years
According to question,
∵ At the time of daughter's birth the age of mother = 28 years
∴ At the time of daughter's birth the age of mother = (9x – 2x) years

$\therefore 9x - 2x = 28$
$7x = 28$
$x = 4$

Age of daughter = 2x = 2 × 4 = 8 years

3. The difference between the age of mother and daughter is 20 years. After 5 years the age of daughters will be half of her present age of mother. What is the age of daughter?

(a) 15 Years (b) 12 Years
(c) 10 Years (d) 8 Years

RRB JE - 30/05/2019 (Shift-I)

Ans : (c) Let the present age of mother is x year and the present age of daughter is y year.
According to question,

$x - y = 20$ ....(i)

After 5 years the age of daughter will be half of the present age of mother

$\Rightarrow \frac{x}{2} = y + 5$

$x - 2y = 10$ ....(ii)

by subtracting equation (i) from equation (ii)

$x - y = 20$
$x - 2y = 10$
$- \quad + \quad -$
$y = 10$

Hence, present age of daughter is 10 years.

4. After 4 years, the total age of the two members of the family will be 64. Four years ago the ratio of their age was 3 : 1. Find the age of the younger member.

(a) 10 (b) 16
(c) 12 (d) 15

RRB JE - 01/06/2019 (Shift-II)

Ans : (b) Let the age of younger member = y year
and age of elder member = x year
First condition-

$x + 4 + y + 4 = 64$
$x + y = 56$ ----(i)

Second condition-

$\frac{x-4}{y-4} = \frac{3}{1}$

$x - 4 = 3y - 12$
$x - 3y = -8$ ----(ii)

from equation (i) and (ii)

$x + y = 56$
$x - 3y = -8$
$- \quad + \quad +$
$4y = 64$
$y = 16$

Hence the age of younger member = y = 16 years

5. The sum and the difference of the ages of two children is 33 and 3 respectively. What is the age of the elder one?

(a) 15 Years (b) 16 Years
(c) 18 Years (d) 24 Years

RRB JE - 01/06/2019 (Shift-II)

Ans : (c) Let the age of elder child = x years
And age of younger child = y years
According to question,
$x + y = 33$ ----(i)
$x - y = 3$ ----(ii)
from equation (i) and (ii)
$x + y = 33$
$x - y = 3$
$2x = 36$
$x = 18$
Hence age of elder child = 18 years

6. The ratio of age of three persons are 4:7:9. 8 years before total age of them were 56. What is present age of the eldest one?
(a) 28 (b) 32
(c) 36 (d) 42
RRB JE - 01/06/2019 (Shift-III)

Ans. (c) Let the present age of first person = 4x years
Present age of second person = 7x years
Present age of third person = 9x years
before 8 years sum of ages of all three persons = 56 years
$4x - 8 + 7x - 8 + 9x - 8 = 56$
$20x = 80$
$x = 4$
Hence present age of eldest person
$= 9x = 9 \times 4 = 36$ years

7. The father's age at the time of his sons' birth was equal to the present age of son. If the age of son is 20, then find the present age of the father.
(a) 39 Years (b) 30 Years
(c) 60 Years (d) 40 Years
RRB JE - 26/06/2019 (Shift-III)

Ans : (d) If present age of son = 20 Years
So presents age of father = (20+20) = 40 Years

8. P is 2 years older than Q, Q is 4 years younger than R. If the sum of their ages is 27 then the age of Q is?
(a) 8 Years (b) 9 Years
(c) 11 Years (d) 7 Years
RRB JE - 27/06/2019 (Shift-III)

Ans : (d) Let the age of Q = x Years
Age of P = (x +2) Years
Age of R = (x + 4) Years
from question–
$(x + x + 2 + x + 4) = 27$
$3x + 6 = 27$
$3x = 21$
$x = 7$
The age of Q is 7 years.

9. Six years ago, the ratio of the ages of two persons P and Q was 3 : 2. After four years, ratio their age will be 8:7. What is the age of P?
(a) 10 Years (b) 12 Years
(c) 14 Years (d) 8 Years
RRB RPF Constable – 19/01/2019 (Shift-II)

Ans : (b)
Let the age of P and Q befor 6 years was 3x and 2x.
Present age of P and Q is (3x+6) year and (2x+6) years
According to question,
$$\frac{3x+10}{2x+10} = \frac{8}{7}$$
$21x + 70 = 16x + 80$
$5x = 10$
$x = 2$
Present age of P = (3x+6) = 3×2+6 = 12 Years

10. The ratio of the age of Deepika and her mother is 3 : 11. After 3 years theirs age ratio becomes 1:3. What is the age of Deepika.
(a) 15 Years (b) 9 Years
(c) 13 Years (d) 11 Years
RRB RPF SI – 11/01/2019 (Shift-I)

Ans. (b) : Let the age of Deepika and her mother is 3x and 11x years respectively.
According to question,
$$\frac{3x+3}{11x+3} = \frac{1}{3}$$
$\Rightarrow 9x+9 = 11x+3$
$\Rightarrow 2x = 6$
$\Rightarrow x = 3$
Therefore age of Deepika = 3×3 = 9 Years

11. The ratio of present ages of X and Y is 3 : 4. Five years ago their age ratio was 5:7. Then what is the present age of Y.
(a) 50 Years (b) 60 Years
(c) 30 Years (d) 40 Years
RRB RPF Constable – 17/01/2019 (Shift-I)

Ans. (d) : Let the present age of X and Y is 3x and 4x years respectively.
The Ratio of age before 5 years = 5 : 7
$$\frac{3x-5}{4x-5} = \frac{5}{7}$$
$21x - 35 = 20x - 25$
$21x - 20x = -25 + 35$
$x = 10$
So present age of Y = 4x = 4 × 10 = 40 Years

12. The ratio of the age of father, mother and daughter is 22:20:9. After 10 years the age ratio of will be 27:25:14. Find the present age of the mother.
(a) 21 (b) 26
(c) 27 (d) 40
RRB Group-D – 07/12/2018 (Shift-III)

Ans : (d) Let the age of father, mother and daughter is 22x, 20x and 9x years respectively.
According to question,-
$$\frac{22x+10}{20x+10} = \frac{27}{25}$$
$550x + 250 = 540x + 270$
$10x = 20$
$x = 2$
The present age of Mother = 20x
$= 20 \times 2 = 40$ Years

13. 6 years ago, the ratio of age of Saina and Sagar was 6 : 5 therefore in the next four years the ratio of their age will be 11:10. What is the present age of Sagar?

(a) 14 Years (b) 16 Years
(c) 12 Years (d) 18 Years

RRB Group-D – 31/10/2018 (Shift-I)

Ans : (b)
Let the age of Saina and Sagar before 6 years
= 6x, 5x Years
The present age of Saina and Sagar = (6x + 6)Years, (5x + 6) Years
According to question,-

$$\frac{6x+6+4}{5x+6+4}=\frac{11}{10}$$

60x + 100 = 55x + 110
5x = 10
x = 2
Present age of Sagar = 5x + 6 = 5 × 2 + 6 = 16 Years

14. S is 7 years younger than R. If their age ratio is 7:9, then what is the age of S.

(a) 16 Years (b) 28 Years
(c) 18 Years (d) 24.5 Years

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (d) Let the age of S = x Years
Age of R = (x + 7) Years
According to question,-

$$\frac{x}{x+7}=\frac{7}{9}$$

$\Rightarrow$ $9x = 7x + 49$
$\Rightarrow$ $2x = 49$
$\Rightarrow$ $x = 24.5$ Years
Therefore Age of S is 24.5 years.

15. The sum of the ages of 6 persons A, B, C, D E and F working in the same company is 105 years. There is a diffeence of 5 years when they are born. What is the age of the eldest one?

(a) 20 Years (b) 25 Years
(c) 30 Years (d) 15 Years

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (c) Let the age of eldest person = x years
∴ Therefore the age of All 6 people is
x, (x –5), (x – 10), (x–15), (x – 20) and (x – 25) Years
According to question,
x + x – 5 + x – 10 + x – 15 + x – 20 + x – 25 = 105
6x – 75 = 105
6x = 180
$\boxed{x = 30}$
Therefore the age of eldest person = 30 years.

16. The present age ratio of Naresh and suparna is 7:3, three years later from now their age ratio will be 2:1. Present age of Naresh is ___ years.

(a) 10.5 (b) 28
(c) 14 (d) 21

RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (d) Let the present age of Naresh = 7 x
Present age of suparna = 3x
Age of Naresh after 3 years = (7x + 3)
Age of suparna after 3 years = (3x + 3)
According to question,

$$\frac{7x+3}{3x+3}=\frac{2}{1}\Rightarrow 7x+3=(6x+6)$$

$\Rightarrow$ x = 6–3
$\Rightarrow$ $\boxed{x = 3}$
Present age of Naresh = 7 × 3 = 21 Years

17. The ratio of the present ages of x and y is 5: 4. Three years later from now, their age ratio will be 11:9. What is the present age of Y.

(a) 26 Years (b) 22 Years
(c) 27 Years (d) 24 Years

RRB Group-D – 25/09/2018 (Shift-II)

Ans : (d)
Let the present age of X and Y = 5x and 4x Years
According to question,

$$\frac{5x+3}{4x+3}=\frac{11}{9}$$

45x + 27 = 44x + 33
x = 33 – 27
x = 6
Present age of Y = 4 × 6 = 24 Years

18. The present ages of S and A are in the ratio of 5: 4 respectively. Therefore after three years, their age ratio will be 11:9 respectively. What is the present age of S in years?

(a) 24 (b) 30
(c) 40 (d) 27

RRB Group-D – 10/10/2018 (Shift-I)

Ans : (b) Let the present age of S = 5x years
Present age of A = 4x Years
from question–

$$\frac{5x+3}{4x+3}=\frac{11}{9}$$

45x + 27 = 44x + 33
x = 6
Therefore present age of S = 5x = 5 × 6 = 30 Years

19. Aman is as young as vinay and he is older than Arun. If the sum of the ages of Arun and Vinay is 40 years, then what is the age of Aman?

(a) 20 Years (b) 22 Years
(c) 25 Years (d) 30 Years

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (a) According to question,
Age of Vinay – age of Aman = age of Aman – age of Arun
Age of Vinay + age of Arun = 2 × age of Aman
2 × Age of Aman = 40
Age of Aman = 20 Years

20. The present age ratio of J and K is 11:6. After 5 years their age ratio will be 12 :7. What is the present age of K.

(a) 30 Years (b) 60 Years
(c) 55 Years (d) 35 Years

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (a) Let the present age of J and K is 11x and 6x years respectively.
According to question,

$$\frac{11x+5}{6x+5}=\frac{12}{7}$$

77x + 35 = 72x + 60
77x – 72x = 60 – 35
5x = 25
x = 5
Therefore, present age of K = 6 × 5 = 30 Years

21. The ratio of present age of Rekha and Rashmi is 7:4. Three years later from now their age ratio will be 8:5. What is the present age of Rashmi (in years)?

(a) 15 (b) 9
(c) 8 (d) 12

RRB Group-D – 12/11/2018 (Shift-I)

Ans. (d) : Let the present age of Rekha and Rashmi is 7x and 4x years.
According to question,

$$\frac{7x+3}{4x+3}=\frac{8}{5}$$

35x + 15 = 32x + 24
3x = 9
x = 3
Therefore, Present age of Rashmi = 4x = 4×3 = 12Years

22. The ratio of present age of Meena and Seena is 4:3. After 6 years age of Meena will be 26. What is the present age of Seena?

(a) 12 Years (b) 19 Years 6 months
(c) 15 Years (d) 21 Years

RRB Group-D – 12/11/2018 (Shift-III)

Ans : (c) Let the present age of Meena = 4x Years
and present age of Seena = 3x Years
According to question,
4x + 6 = 26
4x = 20
x = 5
Present age of Seena = 3x = 3 × 5 = 15 Years

23. When the age of father is 54 years, then the difference between the ages of the two sisters is 4 years. Father is two years older than mother. The younger sister is half of the age of the mother. Find the age of the elder sister.

(a) 26 (b) 27
(c) 29 (d) 30

RRB NTPC 06.04.2016 Shift : 1

Ans : (d) The age of father = 54 years
∴ Age of mother = 54 − 2 = 52 Years

Age of younger sister $=\frac{\text{Age of Mother}}{2}=\frac{52}{2}=26$ year

Age of elder sister – age of younger sister = 4 years
Age of elder sister − 26 = 4
∴ Age of elder sister = 4 + 26 = 30 Years

24. There is a difference of two years in the age of two sisters, while the age of her father is 52 years. Father is two years older than mother. If the age of the elder sister is half of the age of mother, find the age of the younger sister.

(a) 27 (b) 21 (c) 25 (d) 23

RRB NTPC 06.04.2016 Shift : 2

Ans : (d) Let the age of elder sister = x Years
then age of younger sister = (x – 2) Years
Age of father = 52 Years
Age of Mother = 52 -2 =50 Years

$$\frac{\text{Age of mother}}{2}=\text{Age of elder sister}$$

$$\frac{50}{2}=x$$

Elder sister (x) = 25 Years
Younger sister = x – 2 = 25 − 2 = 23 Years

25. Sarika has three children. First is 5 years older than the second one and the second one is 4 years older than the third. The sum of their ages is 22 years. Find the age of the eldest child.

(a) 7 (b) 9 (c) 11 (d) 12

RRB NTPC 29.04.2016 Shift : 1

Ans : (d) Let the age of second child = x Years
then age of third child = (x – 4) Years
and age of first child = (x + 5) Years
According to question,
x – 4 + x + x + 5 = 22
3x + 1 = 22
3x = 21
x = 7
Age of eldest child = x + 5= 7 + 5 = 12 Years

26. Neetu is 10 years elder to Meetu, and Meetu is 7 years elder to Geetu. If the sum of their ages is 48 years, what is Neetu's age (in years)?

(a) 25 (b) 22
(c) 28 (d) 27

RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (a) According to question, where

$$\left\{\begin{array}{l} N \Rightarrow \text{Neetu} \\ M \Rightarrow \text{Meetu} \\ G \Rightarrow \text{Geetu} \end{array}\right\}$$

N = M + 10 .....(i)
M = G + 7 .....(ii)
N + M + G = 48 .....(iii)
from equation (i), (ii) and (iii)–
M + 10 + M + M – 7 = 48
3M = 45
M = 15
Age of Meetu is 15 years,
Age of Neetu = M + 10 = 25 Years

27. Two-thirds of my present age is the same as three-fourths of that of one of my cousins'. My age three years ago will be his age four years hence. My present age is _____ years.

(a) 72 (b) 63
(c) 54 (d) 81

RRB ALP & Tec. (29-08-18 Shift-I)

Ans : (b) If my present age = x year and present age of my cousin = y year
Then according to first condition–

$\frac{2x}{3} = \frac{3y}{4}$

8x – 9y = 0 ..... (i)

According to second condition–

x – 3 = y + 4
x–y = 7
y = x – 7
y = x–7 putting in equation (i)–
8x – 9 (x–7) = 0
8x – 9x + 63 = 0
–x + 63 = 0
x = 63 years

28. The ages difference between A and B is 16 years. If 6 years ago, the elder one was 3 times as old as the younger one, find the present age of the younger between A and B ?

(a) 15 years (b) 11 years
(c) 14 years (d) 12 years

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (c) Let Age of B = x Years
∴ Age of A = (16 + x) Years
According to question, before 6 Years–
3 (x –6) = (16 + x –6)
3x–18 = x + 10
2x = 28
x = 14
Therefore, the age of younger person is 14 years.

29. Present ages of Sai and Sateesh are in the ratio of 5 : 4 respectively. Three years after, the ratio of their ages will be 11 : 9 respectively. What is the present age of Sateesh in years?

(a) 22 (b) 23
(c) 21 (d) 24

RRB ALP & Tec. (10-08-18 Shift-I)

Ans : (d) Let the present age of Sai and Sateesh is 5x years, 4x years respectively.
According to question,

$\frac{5x+3}{4x+3} = \frac{11}{9}$

⇒ 9(5x+3) = 11(4x+3)
⇒ 45x+27 = 44x +33
⇒ 45x – 44x = 33 – 27
⇒ x = 6
Therefore, Present age of Sateesh = 6×4 = 24 Years

Type - 6

30. Neeraj's age is half of suraj's age. If 8 years is subtracted from Neeraj's age and 5 years is added in Suraj's age then Suraj's age will be 5 times more than Neeraj. Suraj's and Neeraj's age was -

(a) 30 Years and 15 Years
(b) 28 Years and 14 Years
(c) 26 Years and 12 Years
(d) 28 Years and 13 Years

RRB RPF SI – 16/01/2019 (Shift-II)

Ans. (d) : Let the present age of suraj is x years, and present age of Neeraj is $\frac{x}{2}$ years.
According to question,

$5\left(\frac{x}{2}-8\right) = (x+5)$

⇒ $\frac{5x}{2} - 40 = x + 5$

⇒ $\frac{5x}{2} - x = 5 + 40$

⇒ $\frac{3x}{2} = 45$

⇒ x = 30 Years
Age of Suraj before 2 years = 30 – 2 = 28 Years
Age of Neeraj before 2 years = 15 – 2 = 13 Years

31. The age ratio of A and B is 3:2. After 10 years the sum of their age will be 80. What is their present age?

(a) 27, 28 (b) 36, 24
(c) 42, 28 (d) 45, 30

RRB JE - 23/05/2019 (Shift-I)

Ans : (b) Let the present age of A = 3x Years
Present age of B = 2x Years
According to question,
3x + 10 + 2x + 10 = 80
5x = 60
x = 12
Present age of A = 3 × 12 = 36 Years
Present age of B = 2 × 12 = 24 Years

32. One year before the age ratio of two sisters was 2:3. Sum of their present age is 12. Find their present age.

(a) 9, 3 (b) 7.5, 4.5
(c) 8, 4 (d) 5, 7

RRB JE - 23/05/2019 (Shift-III)

Ans : (d) Let the present age of both sisters is x years and y years.
x + y = 12 ..... (i)
According to question,

$\frac{x-1}{y-1} = \frac{2}{3}$

⇒ 3x – 3 = 2y – 2

$\Rightarrow 3x - 2y = 1$ ..... (ii)
from equation (i) and (ii)
$[x + y = 12] \times 2$ {multiplying by 2 in eq^n (i)}
$3x - 2y = 1$

$5x = 25$
$x = 5$
putting the value of x in equation (i)
$x + y = 12$
$5 + y = 12$
$y = 7$
Therefore, their present age is = 5 Years and 7 Years

33. The sum and the difference of age ratio of two brothers is 5:1. If the product of their age is 96, then find their age?
(a) 8, 12 (b) 6, 16
(c) 6, 10 (d) 24, 4
RRB JE - 26/06/2019 (Shift-III)

Ans : (a) Let the age of both brothers = x, y Years
According to question,
$$\frac{x+y}{x-y} = \frac{5}{1}$$
$x + y = 5x - 5y$
$6y = 4x$
$$y = \frac{2}{3}x$$
$\because$ $xy = 96$ ....(i)
$$x \times \frac{2}{3}x = 96$$
$$x^2 = 96 \times \frac{3}{2}$$
$x = \sqrt{144}$
$x = 12$
putting the value of x in equation (i)
$12 \times y = 96$
$y = 8$
Therefore, their age is 8, 12.

34. The sum of the age of Sindu and Bindu is 30 years? The product of their age is 221. What is theirs age?
(a) 17 Years, 13 Years
(b) 16 Years, 14 Years
(c) 15 Years, 15 Years
(d) 18 Years, 12 Years
RRB Group-D – 16/10/2018 (Shift-I)

Ans. (a) : Let the age of Sindu is x years and age of Bindu is (30-x) years.
According to question,
$x \times (30 - x) = 221$
$30x - x^2 = 221$
$x^2 - 30x + 221 = 0$
$x^2 - 17x - 13x + 221 = 0$
$x(x - 17) - 13(x - 17) = 0$
$(x - 17)(x - 13) = 0$
If $x - 17 = 0$
x = 17 so another's age = 30-17=13 Years
If x-13=0
x =13 so another's age = 30-13 = 17 Years

35. Sudha's age is twice as of Radha. If 6 years is subtracted from Radha's age and 4 years is added to Sudha's age, then Sudha's age will be four times of Radha's age, Two years before from now, what will be the age of Sudha and Radha?
(a) 30 Years and 14 Years
(b) 26 Years and 12 Years
(c) 34 Years and 20 Years
(d) 30 Years and 16 Years
RRB Group-D – 09/10/2018 (Shift-I)

Ans. (b) : Let the present age of Radha is x years and present age of Sudha is 2x years.
According to question,
$4(x - 6) = (2x + 4)$
$\Rightarrow 4x - 24 = 2x + 4$
$\Rightarrow 4x - 2x = 28$
$\Rightarrow 2x = 28$
$\Rightarrow x = 14$
$\therefore$ age of Sudha before 2 Years $= 2x - 2$
$= 2 \times 14 - 2$
$= 28 - 2 = 26$ Years
and age of Radha before 2 years = 14 – 2 = 12 Years

36. Kiran said to Varun that her brother Abhi is 4 year younger as compared to her. She also told that the sum of their age was 32. Help Varun in finding out the age of Kiran and Abhi.
(a) Kiran's age is 18 years and Abhi's age is 14
(b) Kiran's age is 16 years and Abhi's age is 12
(c) Kiran's age is 12 years and Abhi's age is 16
(d) Kiran's age is 10 years and Abhi's age is 14
RRB Group-D – 11/10/2018 (Shift-III)

Ans : (a) Let the age of Kiran = x Years
Then age of brother Abhi = (x–4) Years
$\because$ $x + x - 4 = 32$
$\therefore$ $2x = 36$, $\Rightarrow x = 18$
Age of Kiran (x) = 18 Years
Age of Abhi = (x – 4) Years = 18 – 4 = 14 Years

37. A man says to his son seven years before I was 7 times of your age and three years after of your age, I will be three times of your age. Find his age.
(a) 42, 12 (b) 52, 12
(c) 40, 14 (d) 20, 42
RRB Group-D – 12/12/2018 (Shift-I)

Ans. (a) Let the age of father is y years and age of son is x years.
According to question,
$7(x-7) = (y-7)$
$7x - 49 = y - 7$
$7x - y = 42$ .....(i)
And
$3(x+3) = (y+3)$
$3x + 9 = y + 3$
$3x - y = -6$ .....(ii)
from equation (i) and (ii)
$7x - y = 42$
$3x - y = -6$
$- \quad + \quad +$
$4x = 48$
$x = 12$
from equation (i)
$7x - y = 42$
$7 \times 12 - y = 42$
$84 - 42 = y$
$y = 42$
therefore age of father = 42 Years
and age of son = 12 Years

38. Ankita is two years younger than Anu. Four years after Anu's age before three years will be twice the age of Ankita. Find the present age of Ankita and Anu.
(a) 14 Years and 16 Years
(b) 15 Years and 17 Years
(c) 12 Years and 14 Years
(d) 13 Years and 15 Years

RRB Group-D – 10/12/2018 (Shift-I)

Ans. (c) : Let the present age of Anu = x Years
Then present age of Ankita = (x – 2) Years
According to question,
$(x+4) = (x-2-3) \times 2$
$x + 4 = (x-5) \times 2$
$x + 4 = 2x - 10$
$x = 14$
Present age of Ankita = x – 2
= 14 – 2 = 12 Years
Therefore, the present age of Ankita and Anu is 12 years and 14 years respectively.

39. The sum of present age of Nitya and Satya is 40 years.The product of their age is 351. What is their present age in years?
(a) 27 and 13 (b) 28 and 12
(c) 25 and 15 (d) 26 and 14

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (a) Let the present age of Nitya = a years
Present age of Satya = b years
According to question, $a + b = 40$ .....(i)
$ab = 351$ .....(ii)
$(a-b)^2 = (a+b)^2 - 4ab$
$= (40)^2 - 4 \times 351$
$= 1600 - 1404$
$a - b = \sqrt{196} = 14$
$a - b = 14$ .....(iii)
from equation (i) and (iii)
$a = 27, b = 13$
∵ Their present age in years is 27 years and 13 years.

40. Father's age is four times the age of the son. In 20 years his age will be twice of his son's age. Find his age?
(a) 36 Years, 9 Years
(b) 44 Years, 11 Years
(c) 40 Years, 10 Years
(d) 60 Years, 15 Years

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (c) : Let the present age of son = x Years
Present age of father = 4x Years
According to question,
$2(x+20) = 4x + 20$
$2x + 40 = 4x + 20$
$40 - 20 = 4x - 2x$
$2x = 20$
$x = 10$ Years
Therefore present age of son = 10 Years
present age of father = 4 × 10 = 40 Years

41. Age ratio of two brothers is 5:8 and the difference of their age is 12 years, find their age.
(a) 20, 32 (b) 16, 28
(c) 18, 30 (d) 22, 34

RRB NTPC 31.03.2016 Shift : 3

Ans : (a) Let the age of both brothers is 5x and 8x years respectively.
from question–
$8x - 5x = 12$
$3x = 12 \Rightarrow x = 4$
Therefore, the age of both brothers is 20 years 32 years.

42. The difference in ages of Sunita and Sheela is 12 years. If 9 years before the elders age was 4 times than, that of younger one, then what is their present age?
(a) 11 and 23 (b) 15 and 27
(c) 13 and 25 (d) 23 and 35

RRB NTPC 18.04.2016 Shift : 2

Ans : (c) Let the present age of sunita = x Years
Present age of sheela = (x – 12) Years
According to question,
$x - 9 = 4(x - 12 - 9)$
$x - 9 = 4(x - 21)$
$x - 9 = 4x - 84$
$4x - x = 84 - 9$
$3x = 75$
$x = 25$
Present age of Sunita = x = 25 Years
Present age of Sheela = x – 12 = 25 – 12 = 13 Years

43. The difference of the age of Vinay and Vijay is 20 years. If 5 years before the elder boys age was 5 times of younger boy. Then what is their present age?

(a) 15 and 35 (b) 5 and 25
(c) 10 and 30 (d) 8 and 28

RRB NTPC 22.04.2016 Shift : 3

Ans : (c) Let the present age of Vijay = x Years
Present age of Vinay = (x+20) Years
According to question,
$x + 20 - 5 = 5(x-5)$
$x + 15 = 5x - 25$
$4x = 40$
$\therefore x = 10$
Therefore, their present age is 10 and 30 years.

Type - 3

44. The present age ratio of mother and daughter is 8:3. After 12 years their age ratio will be 2:1. What is the sum of the present age of mother and daughter.

(a) 66 Years (b) 74 Years
(c) 71 Years (d) 69 Years

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (a) Let the present age of Mother and Daughter is 8x years and 3x years
According to question,
$$\frac{8x+12}{3x+12} = \frac{2}{1}$$
$8x + 12 = 6x + 24$
$2x = 12$
$x = 6$
$\therefore$ Present age of (Mother + daughter) = (8x + 3x) = 11x
= 11 × 6 = 66 Years

45. The age ratio of father and son is 3:1 and the product of their age is 147. Find the sum of their age.

(a) 28 (b) 32
(c) 36 (d) 35

RRB JE - 24/05/2019 (Shift-I)

Ans : (a) Let the age of father = 3x Years
Age of son = x Years
According to question,
$3x \times x = 147$
$x^2 = 49$
$x = 7$
Sum of age of father and son = 3x + x = 4x
= 4 × 7 = 28 Years

46. The age ratio of mother and daughter is 12:5. After 10 years, mother's age will be twice to the age of daughter. What is the sum of their age at present?

(a) 60 (b) 75
(c) 65 (d) 85

RRB JE - 27/05/2019 (Shift-II)

Ans : (d) Let the present age of mother is 12x years and present age of daughter is 5x years.
According to question,
After 10 years the age of mother will be twice to the age of daughter
$12x + 10 = 2 \times (5x + 10)$
$12x + 10 = 10x + 20$
$2x = 10$
$x = 5$
sum of their age at present = 12x + 5x
= 17x
= 17 × 5
= 85 Years

47. A's age is twice of B. Sum of their present age is 60 years. What will be the sum of their age after 5 years?

(a) 70 (b) 80
(c) 65 (d) 75

RRB JE - 31/05/2019 (Shift-I)

Ans : (a) Let the age of B = a Years
Age of A = 2a Years
$2a + a = 60$
$3a = 60$
$a = 20$
Sum of their age after 5 years = 2a + 5 + a + 5
= 2 × 20 + 5 + 20 + 5
= 40 + 30 = 70 Years

48. The difference between the age of A and B is 6. Their ratio is 3:5. Find the sum of their age.

(a) 24 (b) 40
(c) 16 (d) 32

RRB JE - 31/05/2019 (Shift-III)

Ans. (a) Let the age of A = 3x Years
and Age of B = 5x Years
According to question,
$5x - 3x = 6$
$2x = 6$
$x = 3$
Age of A = 3 × 3 = 9 Years
Age of B = 5 × 3 = 15 Years
Sum of their age = 9 + 15 = 24 Years

49. If P's age is twice of Q's age and after 5 years the sum of their age is 70, then find their present age (in years).

(a) 60 (b) 40
(c) 30 (d) 50

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (a) Let the age of Q is x years and Age of P is 2x years.
So According to question,-
$2x + 5 + x + 5 = 70$
$3x = 60$
$x = 20$
Sum of age of P and Q = $2x + x$ = 40 + 20 = 60 Years

50. The present age ratio of P and D is 3 : 4 while the present age ratio of D and A is 5:6. After 15 years from now, the age ratio of P and D will be 4:5. What is the sum of the present age of all three?

(a) 180 Years (b) 183 Years
(c) 177 Years (d) 175 Years

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (c)

P : D and D : A
3 : 4 5 : 6

P : D : A
3 : 4
5 : 6

P:D:A =15 : 20 : 24
Let the present age of P, D and A is 15x, 20x and 24x
Age after 15 years
Age of P = (15x + 15)
Age of D = (20x + 15)
According to question,

$$\frac{15x+15}{20x+15}=\frac{4}{5}$$

75x + 75 = 80x + 60
5x = 15
$\boxed{x = 3}$
Sum of Present age of all three people
= 15 × 3 + 20 × 3 + 24 × 3
= 45 + 60 + 72 = 177 Years

51. The difference between the age of Waqar and Nasima is 8 years. 30 years before they were married then the four times of Waqar's age was equal to 5 times of Nasima's age. Find the sum of their present age.

(a) 130 Years (b) 134 Years
(c) 132 Years (d) 135 Years

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (c) : Let the present age of Nasima = x Year
So present age of Waqar = (x + 8) Year

before 30 Years,
Age of Nasima = (x – 30) Years
Age of Waqar = (x + 8 – 30) Years
= (x –22) Years
According to condition,
4 (x – 22) = 5 (x – 30)
4x – 88 = 5x – 150
5x – 4x = 150 – 88
x = 62
∴ Present age of Nasima = 62 Years
Present age of Waqar = 62 + 8 = 70 Years
Sum of their age = 62 + 70 = 132 Years

52. The present age ratio of X and Y is 3 : 8 while the present age ratio of Y and Z is 4:5. Two years before the age ratio of X and Y was 1:3. The sum of present age of all three is___?

(a) 126 Years (b) 63 Years
(c) 84 Years (d) 105 Years

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (c) The ratio of present age of X and Y is 3:8 and Ratio of present age of Y and Z is 4 : 5.
Ratio of Present age of X:Y:Z is

X : Y : Z
3 : 8
4 : 5
12 : 32 : 40

Let the present age of X and Y is 12x and 32 x respectively.

$$\frac{12x-2}{32x-2}=\frac{1}{3}$$

36x – 6 = 32x – 2
4x = 4
x = 1
Sum of present age of X, Y and Z = 12 + 32 + 40
= 84 Years

53. 28 months before A's age was 3.5 times of B's age 4 months after from now A's age will be 2.5 times of B's age. What is the sum of their present age?

(a) 23 Years
(b) 22 Years 8 month
(c) 23 Years 2 month
(d) 22 Years 10 month

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (b) Let the present age of A is x month and present age of B is y month.
According to question,-
x – 28 = 3.5 (y–28)
x – 3.5y = 28 – 98
x – 3.5 y = –70 .....(i)
After 4 years
x + 4 = 2.5 (y + 4)
x – 2.5y = 10–4
x – 2.5 y = 6 .....(ii)
subtracting equation (ii) from equation (i)
x – 3.5y = -70
x – 2.5y = 6
– + –
– y = -76
y = 76
from equation (ii),
x – 2.5 × 76 = 6
x – 190 = 6 ⇒ x = 196
Sum of present age of A and B
= x + y = 76 + 196 = 272 months
= 22 Years 8 months

54. At present the age ratio of Teertha and Pradeep is 2:3 while the present age ratio of Pradeep and Umar is 4:5. Four years before the age ratio of Teertha and Pradeep was 5: 8. What is the sum of ages of all three at present?

(a) 140 Years (b) 120 Years
(c) 90 Years (d) 105 Years

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (d)

Teertha : Pradeep = 2 : 3
Pradeed : Umar = 4 : 5
Teertha : Pradeep : Umar
2 : 3
4 : 5
8 : 12 : 15

Let the present age of Teertha, Pradeep and Umar is 8x, 12x and 15x years respectively.
before 4 years.
Age of Teertha = (8x–4) Years
Age of Pradeep = (12x–4) Years
According to question,

$$\frac{8x-4}{12x-4}=\frac{5}{8}$$

64x – 32 = 60x – 20
4x = 12
x = 3

The sum of present age of all three people
= 8 × 3 + 12 × 3 + 15 × 3
= 24 + 36 + 45 = 105 Years

55. Chavan's age before 15 years was 4 times of Pawan's age. Pawan's present age 1/3 of Sunil two years before age, while Chavan present age is 3/5 of Sunil present age. What is the sum of age of all three?

(a) 130 Years (b) 120 Years
(c) 128 Years (d) 125 Years

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (d) : Let the present age of sunil = x Years

$\therefore$ present age of chavan $=\frac{3}{5}x$ Years

so present age of pawan $=\frac{(x-2)}{3}$ Years

According to question,-

$$\frac{3}{5}x-15=4\left(\frac{(x-2)}{3}-15\right)$$

$$\frac{3x-75}{5}=4\left(\frac{x-2-45}{3}\right)$$

3(3x –75) = 20 (x – 47)
9x – 225 = 20 x – 940
20x – 9x = 940 – 225
11x = 715

$$x=\frac{715}{11}$$

x = 65

Sum of present age of all three people

$$=x+\frac{3}{5}x+\frac{(x-2)}{3}$$

$$=\frac{15x+9x+5x-10}{15}$$

$$=\frac{29x-10}{15}$$

$$=\frac{29\times65-10}{15}=\frac{1885-10}{15}=\frac{1875}{15}=125\text{ Years}$$

56. 9 months before, Trisha's age was three times of Rishabh's age who is 4 months older than Yawan. After 25 months from now Yavan 20 times age will be equal Trisha 9 times age. What is the sum of present age of her?

(a) 19 Years 11 months (b) 19 Years 9 months
(c) 19 Years 5 months (d) 18 Years 9 months

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (c) Let before 9 months
Age of Yawan = x month
$\therefore$ Age of Rishabh = (x + 4) months
Age of Trisha = (x + 4) × 3 = (3x + 12) months
In present,
Age of Yawan = (x + 9) months
Age of Rishabh = (x + 13) months
Age of Trisha (3x + 21) months
According to question,
After 25 months–
(3x + 46) × 9 = (x + 34) × 20
27x + 414 = 20x + 680
7x = 266
x = 38
Therefore, present age of Yawan = x + 9 = 47 month
Present age of Rishabh = x + 13 = 51 month
Present age of Trisha = 3x + 21 = 135 month
Sum of present age of all people
= 233 months or 19 Year 5 months

57. Ten months ago, Tiyasha's age was 2.5 times of Rishi's age, who is 15 months older than Shravan. 30 months after from now. 16 times of Shrawan's age will be 7 times of Tiyasha's age. What is the sum of the present ages of Tiyasha, Rishi and Shravan?

(a) 30 years 9 month (b) 31 years 9 month
(c) 31 years 3 month (d) 31 years 11 month

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (c) Before 10 months,
Let the age of shravan = x month
$\therefore$ Age of Rishi = (x + 15) months
Age of Tiyasha = (x + 15) × 2.5 = (2.5x + 37.5) months
In present,
Age of Shravan = (x + 10) months
Age of Rishi = (x + 25) months
Age of Tiyasha = (2.5x + 47.5) months
According to question,
After 30 months,

$(x + 10 + 30) \times 16 = (2.5x + 47.5 + 30) \times 7$
$(x + 40) \times 16 = (2.5x + 77.5) \times 7$
$16x + 640 = 17.5x + 542.5$
$1.5x = 97.5$
$x = 65$
Therefore Present age of Shravan = (x+10) = 75 months
Present age of Rishi = (x + 25) = 90 months
Present age of Tiyasha = (2.5x + 47.5) = 210 months
Sum of present age of all people = 75+90+210
= 375 months
375 months = 31 year 3 months

58. Biju's age is 28 years less than three times of Subham's age. Subham's age is 16 years more than Amir's 4/5 age. Kaveri is 10 years younger than Amir and half of the age of Subham. What is the sum of their ages?

(a) 196 Years (b) 182 Years
(c) 180 Years (d) 188 Years

RRB Group-D – 11/12/2018 (Shift-I)

Ans. (b) : Let the age of Shubham = x Years

Age of Kaveri $= \frac{x}{2}$ Years

Age of Amir $= \left(\frac{x}{2} + 10\right)$ Years

Age of Biju = y Years
According to question,
$y = 3x - 28$
$3x - y = 28$ ---------- (i)

$$16 + \left(\frac{x}{2} + 10\right)\frac{4}{5} = x$$

$$16 + \frac{2x}{5} + 8 = x$$

$$24 = x - \frac{2x}{5}$$

$$24 = \frac{5x - 2x}{5}$$

$$24 = \frac{3x}{5}$$

$x = 40$ Years
putting the value of x in equation (i)
$3x - y = 28$
$3 \times 40 - y = 28$
$y = 92$ Years
Age of Shubham = x Years
= 40 Years

Age of Kaveri $= \frac{x}{2} = \frac{40}{2} = 20$ Years

Age of Amir $= \frac{x}{2} + 10 = \frac{40}{2} + 10 = 30$ Years

Age of Biju = y Years = 92 Years
Sum of age of all people
$= 40 + 20 + 30 + 92 = 182$ Years

59. 7 children with the same birth date were born in seven successive years. The sum of the ages of the eldest three children is 93 years. How many years will be the sum of the ages of the youngest three children?

(a) 81 (b) 90
(c) 87 (d) 84

RRB Group-D – 12/11/2018 (Shift-II)

Ans : (a) Age of all seven children Respectively = x, (x + 1), (x + 2), (x + 3), (x + 4), (x + 5), (x + 6) Years
Sum of age of eldest three children = 93
$(x + 4) + (x + 5) + (x + 6) = 93$
$3x + 15 = 93$
$3x = 93 - 15$
$3x = 78$
$\boxed{x = 26}$
So,
Sum of age of youngest three children = x+(x+1)+(x+2)
= 3x + 3
putting the value of x,
$= 3 \times 26 + 3 = 78 + 3 = 81$ Years

60. Seven children with the same birth date were born in 7 successive years. The sum of the ages of the eldest three children is 63 years. How many years will be the sum of the ages of the youngest three children?

(a) 57 (b) 54
(c) 51 (d) 60

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (c) : Let the age of all seven children = x, (x + 1), (x + 2), (x + 3), (x + 4), (x + 5), (x + 6) Years
Sum of age of eldest three children
= (x+4)+(x+5)+(x+6)
$63 = 3x + 15$
$3x = 48$
$x = 16$
Sum of age youngest three children
$= x + (x + 1) + (x + 2) = 3x + 3$
putting the value of x,
$= 16 \times 3 + 3 = 51$ Years

61. The difference between Peter and Preeti's ages is 5 years. When they married each other 35 years ago, 4 times of Peter's age was the same as 5 times the age of Preeti. What is the sum of their present age?

(a) 105 years (b) 110 years
(c) 115 years (d) 112 years

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (c) If the present age of peter is x years and age of Preeti is y years then,
$x - y = 5$ .....(i)
According to question,
Before 35 years,
$4(x - 35) = 5(y - 35)$
$4x - 140 = 5y - 175$
$4x - 5y = -35$.....(ii)

by multiplying 4 in equation (i) and on solving (ii)
4x–4y=20
4x–5y=–35
y=55 Years
putting y = 55 in equation (i),
x – y = 5
x = 55 + 5 = 60 Years
sum of present age of both = x+y = 60+55 = 115 years

62. The difference between Charles' and Shriya's age is 6 years. When they married each other 30 years ago, 4 times of Charles's age was the same as 5 times the age of Shriya. What is the sum of their present age?
(a) 112 years (b) 114 years
(c) 115 years (d) 110 years
RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (b) Let the age of Shriya = x years
so charles age will be = (x+6) Years
According to question,
(x–30) × 5 = (x+6–30) × 4
5x–150 = 4x–96
x = 150–96
x = 54 Years
∴ Age of Charles = x+6 = 54+6 = 60 Years
Sum of age of both = 60+54 =114 Years

Type - 6

63. If the present age of P is 15 years and after 6 years the age of Q will become 26 years, then what is the ratio of their present age?
(a) 4 : 1 (b) 2 : 3
(c) 2 : 1 (d) 3 : 4
RRB RPF Constable – 25/01/2019 (Shift-III)

Ans. (d) : Present age of P = 15 Years
Let present age of Q = x Years
According to question,-
x + 6 = 26
x = 20 Years
Present age of Q = 20 Years
Ratio of present age of P and Q = 15 : 20 = 3 : 4

64. The age of A, B and C are in the ratio of 2:4:5 and the sum of their age is 77. Find the ratio of the age of A and B after 10 years.
(a) 10 : 17 (b) 12 : 19
(c) 13 : 18 (d) 11 : 14
RRB JE-24/05/2019 (Shift-I)

Ans : (b) Let the age of A = 2x Years
Age of B = 4x Years
Age of C = 5x Years
According to question,
2x + 4x + 5x = 77
11x = 77
x = 7
∴ Ratio of age of A and B after 10 years
= 2x + 10 : 4x + 10
= 2 × 7 + 10 : 4 × 7 + 10
= 24 : 38
= 12 : 19

65. The daughter's age is one-fourth of her father's age. Five years later, she will be one third of her father's age. Later, after 5 years, what will be the ratio of her and her father's age?
(a) 2 : 5 (b) 3 : 5
(c) 3 : 4 (d) 5 : 2
RRB JE - 27/05/2019 (Shift-I)

Ans : (a) Let the present age of father = x Years
∴ present age of daughter = $\frac{x}{4}$ Years
According to question,
$$\frac{1}{3}(x+5)=\left(\frac{x}{4}+5\right)$$
$$\frac{x}{3}+\frac{5}{3}=\frac{x}{4}+5$$
$$\frac{x}{12}=\frac{10}{3}$$
x = 40 Years (Age of father)
Age of daughter = $\frac{x}{4}=\frac{40}{4}$ = 10 Years
Age of both after 10 years =
40 + 10 = 50 Years and 10 + 10 = 20 Years
Age of daughter : age of father = 20 : 50
= $\boxed{2:5}$

66. Seven years ago, the age ratio of Ajit and Ganesh was 5:7. If the product of their present age is 616, then find the ratio of their present age.
(a) 15: 17 (b) 17: 15
(c) 12: 13 (d) 11: 14
RRB JE - 28/05/2019 (Shift-I)

Ans : (d) A G {A = Ajit
5x 7x (before 7 years) G = Ganesh}
↓ ↓
(5x+7) (7x+7) (present age)
According to question,-
(5x+7)(7x+7) = 616
$35x^2 + 35x + 49x + 49 = 616$
$35x^2 + 84x - 567 = 0$
$5x^2 + 12x - 81 = 0$
$5x^2 + 27x - 15x - 81 = 0$
x(5x+27) – 3(5x+27) = 0
(5x+27)(x–3) = 0
x = 3
Ratio of present age = $\frac{5x+7}{7x+7}=\frac{5\times3+7}{7\times3+7}=\frac{22}{28}=\boxed{\frac{11}{14}}$

67. The ages of A, B and C are in the ratio of 2:3:8. The sum of their ages is 108. What will be the ratio of their ages after 12 years?

(a) 2 : 5 : 6 (b) 1 : 2 : 3
(c) 1 : 3 : 5 (d) 3 : 4 : 5

RRB JE - 29/05/2019 (Shift-III)

Ans : (d) Let the age of A, B and C = 2x, 3x, 4x Years
from question–
2x + 3x + 4x = 108
9x = 108
$\boxed{x = 12}$
Ratio of their age after 12 years
= 2x + 12 : 3x + 12 : 4x + 12
= 24 + 12 : 36 + 12 : 48 + 12
= 36 : 48 : 60
Ratio of their ages = 3 : 4 : 5

68. The father's age is 3 times more than his son. After 8 years his age become 2.5 times of sons age what will be the age ratio of his and his son after 8 years?

(a) 5 : 2 (b) 3 : 1
(c) 2 : 1 (d) 11 : 4

RRB JE - 29/05/2019 (Shift-III)

Ans : (a) Let the age of son = x Years
Age of father = 3x Years
Age of son after 8 years = (x + 8)
Age of father after 8 years = (3x + 8)
from question–
3x + 8 = 2.5 (x+8)
3x +8 = 2.5x + 20
0.5x = 12
$x = \frac{12}{0.5}$
$\boxed{x = 24}$
Age of father after 8 years = 3x + 8
= 3 × 24 + 8 = 72 + 8 = 80
Age of son after 8 years = x + 8 = 24 + 8 = 32
Ratio of their ages = 80 : 32 = 5 : 2

69. Before ten years, mother's age was three times her son's age. After 10 years her age will become twice of the age of son. Find the ratio of their present age.

(a) 7 : 4 (b) 4 : 3
(c) 7 : 3 (d) 3 : 2

RRB JE - 30/05/2019 (Shift-II)

Ans : (c) Let the age of son before 10 years = x Years
∴ age of mother before 10 years = 3x Years
present age of son = (x + 10) Years
present age of mother = (3x + 10) Years
According to question,
3x + 10 + 10 = 2 (x + 10 + 10)
3x + 20 = 2x + 40
x = 20
∴ Ratio of present age of both $= \frac{3x+10}{x+10}$
$= \frac{3\times 20+10}{20+10} = \frac{70}{30} = 7 : 3$

70. Before four years, the ratio of age of A and B was 2:1. After four years, this ratio will become 3:2. Find the ratio of their present age.

(a) 5 : 2 (b) 2 : 5
(c) 5 : 4 (d) 5 : 3

RRB JE - 28/05/2019 (Shift-II)

Ans : (d) Let the age of B before 4 years = x years
and age of A = 2x Years
Present age of A = (2x + 4) Years
Present age of B = (x + 4) Years
According to question,
After 4 years,
$\frac{2x+4+4}{x+4+4} = \frac{3}{2}$
$\frac{2x+8}{x+8} = \frac{3}{2}$
4x + 16 = 3x + 24
x = 8
Present age of A = 2x + 4
= 2 × 8 + 4 = 20 Years
Present age of B = x + 4
= 8 + 4 = 12 Years
Ratio of present age = A : B = 20 : 12 = 5 : 3

71. Pooja's age before 4 years is equal to Deepa's age after 4 years. The ratio of Pooja's age before 4 years and Deepa's age after 4 years is 3:1. What is the ratio of their present age?

(a) 5 : 7 (b) 5 : 2
(c) 4 : 7 (d) 5 : 3

RRB JE - 31/05/2019 (Shift-I)

Ans : (d) Let the present age of Pooja = x years
and present age of Deepa = y Years
First condition,
x – 4 = y + 4
x – y = 8 -----(i)
Second condition,
$\frac{x+4}{y-4} = \frac{3}{1}$
x + 4 = 3y – 12
x - 3y = – 16 -----(ii)
from equation (i) and (ii)-
x – y = 8
x – 3y = –16
– + +
2y = 24
y = 12
putting the value of y in equation (i)
x = 20
Therefore, ratio of present age of pooja and deepa
= 20 : 12 = 5 : 3

72. The present age ratio of P and Q is 6:7. Q is 4 years older than P. The ratio of their age after 4 years will be–

(a) 7 : 9 (b) 7 : 8
(c) 5 : 8 (d) 8 : 3

RRB JE - 02/06/2019 (Shift-III)

Ans : (b) Let the present age of P = x Years
Present age of Q = (x + 4) Years
According to question,

$$\frac{x}{x+4}=\frac{6}{7}$$

7x = 6x + 24
x = 24
Present age of P = 24 Years
Present age of Q = 24 + 4 = 28 Years
Ratio of age of P and Q after 4 years–

$$\frac{24+4}{28+4}=\frac{28}{32}=\frac{7}{8}=7:8$$

73. Before five years, the ratio of age of P and Q was 7:9 after 10 years this ratio will be 5:6. Find the ratio of their age at present.

(a) 6 : 7 (b) 5 : 4
(c) 2 : 1 (d) 4 : 5

RRB JE - 01/06/2019 (Shift-I)

Ans : (d) Let the age of P before 5 years = 7x years
Age of Q = 9x Years
Present age of P = (7x + 5) Years
Present age of Q = (9x + 5) Years
Age of P after 10 years = (7x + 5 + 10) Years
= (7x + 15) Years
Age of Q after 10 years = (9x + 5 + 10) Years
= (9x + 15) Years
According to question,

$$\frac{7x+15}{9x+15}=\frac{5}{6}$$

42x + 90 = 45x + 75
3x = 15
x = 5
Present age of P = 7x + 5
= 7 × 5 + 5
= 40
Present age of Q = 9x + 5
= 9 × 5 + 5
= 50
The ratio of present age–
P : Q = 40 : 50
= 4 : 5

74. Y years before Mauma age was 1/5 of Saumi. Y years after from today Mauma age will be 1/4 of Saumi age. What is the ratio of present age of Mauma and Saumi?

(a) 16 : 25 (b) 4 : 5
(c) 7 : 31 (d) 2 : 9

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (c) Let the present age of Mauma = a Years
Present age of Saumi = b Years
before Y years,

$$a-y=(b-y)\frac{1}{5}$$

5a – 5y = b – y
5a – b = 4y ...(i)
After Y years,

$$a+y=(b+y)\frac{1}{4}$$

4a + 4y = b + y
4a – b = –3y ...(ii)
by solving equatin (i) and (ii)
5a – b = 4y
4a – b = –3y
– + +
a = 7y
Therefore,
5a – b = 4y
5 × 7y – b = 4y
b = 35y – 4y
b = 31y
a : b = 7y : 31y
Therefore: a : b =7 : 31

75. Before y years Juthika's age was one fourth of Ruby's age. After y years from today Juthika age will be 1/3 of Ruby's age. What will be the ratio of present age of Juthika and Ruby?

(a) 5 : 17 (b) 9 : 16
(c) 3 : 4 (d) 2 : 7

RRB Group-D – 27/11/2018 (Shift-III)

Ans. (a) Let the age of Ruby before y years = x years
Age of Juthika $=\frac{x}{4}$ years
present age of Ruby = (x + y)
and present age of Juthika $=\left(\frac{x}{4}+y\right)$
According to question,

$$\left(\frac{x}{4}+y\right)+y=\frac{1}{3}[(x+y)+y]$$

$$\frac{x+4y+4y}{4}=\frac{x+2y}{3}$$

3x + 12y + 12y = 4x + 8y
3x – 4x = 8y – 12y – 12y
x = 16y
∴ present age of Ruby = (16y + y) = 17y
and present age of juthika $=\left(\frac{16y}{4}+y\right)=5y$
Therefore, Ratio of present age of Juthika and Ruby-
5y : 17y = 5 : 17

76. The ratio of the present age of father and son is 2:1. The product of their age is 200, what will be the ratio of their ages after 5 years?

(a) 9 : 5 (b) 10 : 7

(c) 9 : 2 (d) 5 : 3

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (d) Let the present age of father = 2x Years
present age of son = x Years
According to question,
$2x \times x = 200$
$2x^2 = 200$
$x^2 = 100$
$x = 10$
Ratio of their ages after 5 years–
$$\frac{2x+5}{x+5} = \frac{2\times10+5}{10+5} = \frac{25}{15} = 5:3$$

77. Before 10 years, father's age was three times more than his daughter's age. After 10 years, the father's age will be more than double of his daughter's age. What is the ratio of their present age?
(a) 3 : 1 (b) 7 : 3
(c) 5 : 2 (d) 4 : 7

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (b) :

	Age of daughter	age of father
before 10 years→	x Years (let)	3x Years
At present →	(x+ 10)years	(3x + 10) years

According to question,-
$2(x + 20) = (3x + 20)$
$2x + 40 = 3x + 20$
$x = 20$
Present age of daughter = 20 + 10 = 30 Years
Present age of father = 3 × 20 + 10 = 70 Years
∴Required ratio = 70 : 30 = 7 : 3

78. 10 years ago mother's age was three times her son's age. After 10 years the mother's age will be twice the age of his son. What is the ratio of their present age?
(a) 7 : 3 (b) 12 : 5
(c) 4 : 2 (d) 6 : 1

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (a) Let the present age of son is y years and present age of mother is x years.
So according to question,
$(x - 10) = 3(y - 10)$
$x - 10 = 3y - 30$
$x - 3y = -30 + 10$
$x - 3y = -20$ --------(i)
Their age after 10 years-
$(x + 10) = 2(y + 10)$
$x + 10 = 2y + 20$
$x - 2y = 20 - 10$
$x - 2y = 10$ --------- (ii)
By solving equation (i) and (ii),
$x = 70$
$y = 30$
$$\text{RequiredRatio} = \frac{x}{y} = \frac{70}{30} = 7:3$$

79. Before 10 years Murali's age was three times of the age of his son Satish. Ten years later, Murali's age will be two times that of Satish. What is the ratio of their present age?
(a) 9 : 2 (b) 13: 4
(c) 5 : 2 (d) 7: 3

RRB Group-D – 23/09/2018 (Shift-II)

Ans : (d) Let the present age of Murli is x years and present age of his son Satish is y years.
According to question,
Before 10 years–
$(x-10) = 3(y-10)$
$x - 10 = 3y - 30$
$x - 3y = -20$ ----- (i)
After 10 years, $x + 10 = 2(y + 10)$
$x - 2y = 10$ ----(ii)
On solving both equation-
$x = 70$ and $y = 30$
Ratio of present age of murli and satish
$$= \frac{x}{y} = \frac{70}{30} = \frac{7}{3} = 7:3$$

80. The age ratio of Jay and Jog is 5:2. The sum of their age is 63. What will be their age ratio after 9 years?
(a) 5:2 (b) 2:1
(c) 3:2 (d) 4:3

RRB NTPC 03.04.2016 Shift : 1

Ans : (b) Let the present age of Jay and Jog is 5x and 2x respectively.
According to question,
$5x + 2x = 63$
$7x = 63$
$x = 9$
∴ Ratio of age after 9 years = (5×9+9) : (2×9+9)
= 54 : 27
= 2 : 1

81. The present ratio of the age of Seema and Reema is 2:3. Seema is 6 years younger than Reema. After 6 years the age ratio of Seema and Reema will be.
(a) 2:3 (b) 2:7
(c) 3:4 (d) 7:8

RRB NTPC 22.04.2016 Shift : 1

Ans : (c) Given–
Present age of Seema and Reema is 2x and 3x respectively.
According to question,-
$3x - 2x = 6$
$x = 6$ Years
Therefore Present age of Seema and Reema = 2×6, 6×3
= 12 Years, 18 Years
Ratio of age of Seema and Reema after 6 years–
= (12+6) : (18 + 6)
= 18 : 24 = 3 : 4

Type - 5

82. The average age of 4 persons is 42 years. If their age are in the ratio of 1:3:4:6 respectively. Find the difference between the age of the youngest and the eldest person.

(a) 61 Years (b) 60 Years
(c) 70 Years (d) 59 Years

RRB JE - 26/05/2019 (Shift-II)

Ans : (b) Let the age of all 4 people = x, 3x, 4x and 6x years
Average age of 4 people = 42 years
Sum of age of all 4 people = 42 × 4 = 168 Years

$x + 3x + 4x + 6x = 168$
$14x = 168$
$x = \frac{168}{14}$
$\boxed{x = 12}$

Age of eldest person = 6x
= 6 × 12
= 72 Years
Age of youngest person = 12 Years
Age difference between both = 72 – 12 = 60 Years

83. When Asha was born at that time her father's age was 38. When her brother was born. Who is 4 year younger than her, her mother's age was 36 years. Find the age difference of their parents.

(a) 8 Years (b) 6 Years
(c) 4 Years (d) 2 Years

RRB JE - 28/05/2019 (Shift-II)

Ans : (b) Let the present age of Asha = x Years
Present age of father = (x + 38) Years
Present age of borther = (x – 4) Years
Present age of mother = (36 + x – 4) Years
= (32 + x) Years
age difference between father and mother
= (x+38) – (32+x) = 6 years

84. When the age of mother was 43 years, there was a difference of 21 years between the age of mother and son. If the father is 3 years older than mother, when the father's age will be 50 years, what will be the difference between the age of father and son?

(a) 21 (b) 22
(c) 23 (d) 24

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (d) Age of Mother = 43 Years,
Age of Son = 43–21 = 22 Years
Age of Father = 46 Years
Age difference b/w father and son after 4 years=
= 50 – 26 = 24 Years

85. Monica's father's age was 38 years old when Monica was born, while her mother age was 36 when her four year old younger sister was born. What is the difference between the age of their parents?

(a) 2 Years (b) 8 Years
(c) 4 Years (d) 6 Years

RRB Group-D – 28/09/2018 (Shift-II)

Ans. (d) : Monica's father's age when Monica's 4 years younger sister was born = 38 + 4 = 42 Years
Age of her mother at that time = 36 Years
∴ Age difference of mother and father = 42-36 = 6 Years

86. The ratio of the present age of X and Y is 2:1. After 14 years their age ratio will be 29:18. What is the difference between the present age of X and Y.

(a) 22 Years (b) 11 Years
(c) 9 Years (d) 13 Years

RRB NTPC 19.04.2016 Shift : 3

Ans : (a) Let the present age of X and Y is 2x and x respectively.
According to question,

$\frac{2x+14}{x+14} = \frac{29}{18}$
$36x + 252 = 29x + 406$
$7x = 154$
$x = 22$

∴ Age difference = 2x – x = x = 22 Years

Type - 6

87. Tom's father is three times older than Tom. 10 years ago, Tom's father was 7 times of his age. How old is Tom?

(a) 15 Years (b) 16 Years
(c) 14 Years (d) 17 Years

RRB RPF SI – 11/01/2019 (Shift-III)

Ans : (a) Let the present age of Tom = x years
Then present age of Tom's father = 3x Years
According to question,

$7(x - 10) = (3x - 10)$
$7x - 70 = 3x - 10$
$4x = 60,\ x = 15$ Years

Therefore, Present age of Tom (x) = 15 Years

88. Rajan was married 8 years ago. Then he was 5/6 of his present age. At the time of his marriage, his sister was 10 years younger than him. What is the present age of sister?

(a) 38 (b) 32
(c) 26 (d) 40

RRB JE - 22/05/2019 (Shift-I)

Ans : (a) Let the present age of Rajan = x Years
then present age of sister = (x – 10) Years
According to question,

$x - 8 = \frac{5}{6}x$

$x - \frac{5}{6}x = 8$

$\frac{x}{6} = 8$

x = 48 Years

Therefore present age of sister = 48 – 10 = 38 Years

89. Raja is three times older than Arun. Three years ago, he was four times older than Arun. How old is Raja now?

(a) 6 (b) 15
(c) 27 (d) 12

RRB JE - 23/05/2019 (Shift-I)

Ans : (c) Let the age of Raja = 3x Years

Age of Arun = x Years

from question–

(3x –3) = 4 (x–3)

3x – 3 = 4x – 12

$\boxed{x = 9}$

Present age of Raja = 3 × 9 = 27 Years

90. Present age of a son is 2/5 his mother's age. 8 years later her age will be half of her mother's age. What is his mother's present age?

(a) 36 (b) 42
(c) 40 (d) 50

RRB JE - 26/05/2019 (Shift-III)

Ans : (c) Let The present age of mother = x Years

Present age of son $= \frac{2x}{5}$ years

∴ from question,

$\frac{2x}{5} + 8 = \frac{x+8}{2}$

$\frac{2x+40}{5} = \frac{x+8}{2}$

4x + 80 = 5x + 40

x = 40 Years

Present age of Mother = 40 Years

91. Three times of my age before three years, Subtracting from three times of my age after three years, then my present age is found then find my present age.

(a) 21 Years (b) 15 Years
(c) 24 Years (d) 18 Years

RRB RPF SI – 06/01/2019 (Shift-II)

Ans. : (d) Let My present age is x years.

from question–

3 (x+3) – 3 (x–3) = x

3x + 9 – 3x + 9 = x

x = 18 Years

92. The ratio of age of a brother and sister is 4:3. Afer 3 years the age of the sister will be double of her present age. What is present age of brother?

(a) 4 Years (b) 12 Years
(c) 6 Years (d) 8 Years

RRB JE - 29/05/2019 (Shift-I)

Ans : (a)

Let the present age of brother and sister is 4x, 3x years respectively.

Age of sister after 3 years = (3x + 3) Years

According to question,

(3x + 3) = 3x × 2

3x + 3 = 6x

3 = 3x

$\boxed{x = 1}$

Present age of brother = 4x = 4 × 1 = 4 Years

93. The sum of the age of father and son is 45 years. Five years ago, the product of their age was four times that of his father age at that time. What is the present age of father?

(a) 36 (b) 42
(c) 28 (d) 60

RRB JE - 27/06/2019 (Shift-III)

Ans : (a) Let the present age of son = x Years

present age of father = (45 – x) Years

Age of son before 5 years = (x – 5) Years

Age of father before 5 years = (40 – x) Years

According to question,-

(x–5) (40 – x) = 4(40 – x)

x – 5 = 4

x = 9

Present age of father = 45 – x = 45 – 9 = 36 Years

94. A father is 5 times older than his son. Five years ago, he was six times older than his son. Find the age of his son.

(a) 32 Years (b) 35 Years
(c) 25 Years (d) 28 Years

RRB JE - 27/06/2019 (Shift-III)

Ans : (c) Let the present age of son = x years

Present age of father = 5x Years

Age of son before 5 years = x – 5

Age of father before 5 years = 5x – 5

from question–

5x – 5 = 6(x –5)

5x – 5 = 6x – 30

x = 25

Therefore the age of son will be 25 years.

95. A father's age is three times of his son's age and the son's age is 3/8 of his mother's age. If the difference between his mother's age and the father's age is 4 years, then find the age of the son.

(a) 10 Years (b) 9 Years
(c) 11 Years (d) 12 Years

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans : (d) Let the age of son = x years

$\therefore$ Age of father = 3x Years

and age of mother $= \frac{8}{3}x$ Years

According to question,-

$$3x - \frac{8}{3}x = 4$$

$$9x - 8x = 12$$

$$x = 12 \text{ Years}$$

Therefore Age of son = 12 Years

96. 17 years later from today chetna's age will be twice as Mahim's age. Before 5 years from today Mahim's age was one year less than 1/3 part of chetna's age. What is the present age of chetna?

(a) 65 Years (b) 63 Years
(c) 67 Years (d) 61 Years

RRB RPF SI – 05/01/2019 (Shift-II)

Ans : (a) Before 5 years-

Age of Chetna = x Years

Age of Mahima $= \frac{x}{3} - 1 = \left(\frac{x-3}{3}\right)$ Years

After 17 years,

$$x + 5 + 17 = \left(\frac{x-3}{3} + 5 + 17\right) \times 2$$

$$x + 22 = \left(\frac{x-3}{3} + 22\right) \times 2$$

$$x + 22 = \frac{2x-6}{3} + 44$$

$$x - \frac{2x-6}{3} = 22$$

$$3x - 2x - 6 = 66$$

$$x = 60$$

Present age of Chetna = 60 + 5 = 65 Years

97. The sum of the present age of the two cousins is 54. Before 11 years, the elder brother was three times old as the younger one. What is the present age of the eldest brother.

(a) 36 Years (b) 35 Years
(c) 32 Years (d) 34 Years

RRB RPF SI – 13/01/2019 (Shift-II)

Ans : (b) Let the present age of first brother = x years (elder brother)

then present age of second brother = (54–x) years

Age of both before 11 years would be–

(x–11)years, (54–x–11)years or (43–x)years respectively.

According to condition-

$x - 11 = (43 - x) \times 3$

$x - 11 = 129 - 3x$

$4x = 140$

$x = 35$ years

Therefore, present age of elder brother will be 35 years.

98. A's present age is 9 years more than B. After 10 years, A's age will be double of B's age, as of 10 years ago. What is the present age of A?

(a) 48 (b) 39
(c) 36 (d) 23

RRB JE - 30/05/2019 (Shift-II)

Ans : (a) Let the present age of B = x Years

present age of A = (x + 9) Years

According to question,-

$$x + 9 + 10 = 2(x - 10)$$

$$x + 19 = 2x - 20$$

$$x = 39$$

Therefore present age of A = x + 9

= 39 + 9 = 48 years

99. My age is 5 times more than my son's age. In six years, my age will be three times of my son's age. After two years what will be ours age.

(a) 30 Years and 10 Years
(b) 32 Years and 8 Years
(c) 26 Years and 12 Years
(d) 28 Years and 14 Years

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (b) : Let the present age of son = x Years

$\therefore$ Present age of father = 5x Years

According to question,-

$(5x + 6) = (x + 6) \times 3$

$5x + 6 = 3x + 18$

$2x = 12$

$x = 6$

present age of father $\rightarrow$ 5x= 5 × 6 = 30 Years

Age of father after 2 years = 30 + 2 =32 Years

Age of son after 2 years = 6 +2

= 8 Years

100. Pinaki is 9 years younger than Bhaswati. After thirteen years Bhaswati's age will be 1.2 times of Pinaki's age. Find present age of Pinaki.

(a) 28 years (b) 32 years
(c) 30 years (d) 33 years

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (b) Let the present age of Bhasvati = x Years

Present age of Pinaki = (x–9) Years

Age of Bhaswati after 13 years = (x+13) Years

Age of Pinaki after 13 years = (x–9+13) = (x+4)Years

According to question,

$$(x+13) = 1.2(x+4)$$

$$x+13 = 1.2x+4.8$$

$$0.2x = 8.2$$

$$x = \frac{8.2}{0.2} = 41$$

Therefore, Present age of Pinaki = (x–9) = (41–9) = 32 Years

101. Priyamvad's present age is five years more than the twice of his cousin Ritika. 16 years later from today. Priyamvads age will be 150% more than Ritika. At present what is the age of Priyamvads (in years)?

(a) 33 (b) 23
(c) 26 (d) 17

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (d) : Let the age of Ritika = x Years
Present age of Priyamvad = (2x + 5) Years
According to question,-

$$\left(1+\frac{50}{100}\right)(x+16)=2x+5+16$$

$$\frac{3}{2}(x+16)=2x+21$$

$3x + 48 = 4x + 42$

$\boxed{x = 6}$

Present age of Priyamvad = 2x+5 = 2×6+5 = 17 Years

102. 3/7th of my present age is same as 4/5th of one of my cousin's age my age before 3 years was equal to his age after 10 years. What is my present age?

(a) 42 (b) 35
(c) 21 (d) 28

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (d) : Let my present age = x years
and present age of cousin = y Years
According to first condition -

$$x\times\frac{3}{7}=y\times\frac{4}{5}$$

$15x = 28y$ .....(i)

∴ My age before 3 years = (x-3) Years

∴ The age of cousin after 10 years from now.
= (y+10) Years

According to second condition-

$(y+10)=(x-3)$

$x = y+13$

Putting x = y + 13 in equation (i)

$15(y+13)=28y$

$15y+195=28y$

$13y=195$

or y=15 Years

Now my present age = x = 15+13= 28 years

103. Shan's present age is 4 yrs less than 1.6 times of Udhalaks age. Before 26 yrs Udhalaks age was one years less than half of the Shan's age. What is the present age of Shan?

(a) 68 (b) 84
(c) 76 (d) 60

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (c) Let the present age of Udhlak = x Years
∴ Present age of Shan = (1.6x - 4) Years
∴ Age of Udhlak before 26 years = (x - 26)
∴ Age of Shan before 26 years = 1.6x - 4 -26
= (1.6x - 30)

According to question,-

$$\frac{1.6x-30}{2}-1=x-26$$

1.6x - 30 -2 = 2x -52
0.4x = 52 - 32
0.4x = 20

$$x=\frac{20\times10}{4}$$

x = 50

∴ Present age of Shan
50 × 1.6 - 4 = 80 - 4 = 76 Years

104. 18 years ago, Cyrus was 2.5 times older than Nikhil. The sum of their present age is 92 years. What is the present age of Nihkil?

(a) 31 Years (b) 32 Years
(c) 34 Years (d) 33 Years

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (c) Let the age of nikhil before 18 years = x Years
Age of Cyrus before 18 years = 2.5x Years
Present age of Nikhil = (x + 18) Years
Present age of Cyrus = (2.5x + 18) Years
According to question,-

(x + 18) + (2.5x + 18) = 92
3.5x + 36 = 92
3.5x = 92 – 36
3.5x = 56

$$x=\frac{56}{3.5}$$

$$x=\frac{560}{35}$$

$\boxed{x = 16}$

Present age of Nikhil = x +18 = 16 + 18 = 34 years

105. Robin's father is 2.8 times older than his. Six years ago, age of Robin's father was 4 times of his age. What is Robin's present age?

(a) 16 Years (b) 15 Years
(c) 14 Years (d) 17 Years

RRB Group-D – 24/09/2018 (Shift-II)

Ans : (b) Let the present age of Robin = x years
Present age of Robin's father = 2.8x years
Age of Robin before 6 years = (x – 6) Years
Age of father before 6 years = (2.8x – 6) Years
According to question,-

(2.8x – 6) = 4 (x – 6)
2.8 x – 6 = 4x – 24
2.8 x – 4x = – 24 + 6
–1.2x = –18

$$x=\frac{180}{12}$$

x = 15

Therefore age of Robin = 15 Years

106. After three years from now, Panna's age will be four years less than three times of Eugene's age. The sum of their present age is 54 years. What is the present age of Panna?

(a) 36 Years (b) 43 Years
(c) 39 Years (d) 41 Years

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (d) Let the present age of Panna = x Years
$\therefore$ Present age of Eugene = $(54 - x)$ Years
after 3 years,
Age of Panna = $(x + 3)$ Years
Age of Eugene = $(54 - x + 3)$
= $(57 - x)$ Years
According to question,
$3(57 - x) - 4 = x + 3$
$171 - 3x - 4 = x + 3$
$4x = 164$
$x = 41$ Years
Therefore present age of Panna = 41 Years

107. The total age of mother and daughter is 50 years. After 5 years the mother's age will be 4 times more than the daughter's age. Find the present age of the daughter?

(a) 12 years (b) 10 years
(c) 7 years (d) 15 years

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (c) Let the present age of Daughter = x Years
then present age of mother = (50 – x) Years
Age of Mother after 5 years = (50 – x + 5) Years
= (55 – x) Years
Age of Daughter after 5 years = (x + 5) Years
So, according to condition—
$(55 - x) = 4(x + 5)$
$55 - x = 4x + 20$
$55 - 20 = 4x + x \Rightarrow 5x = 35$
$\Rightarrow x = 7$
Therefore present age of Daughter = 7 years

108. Anish is three times older than as old as his son and the sum of their age is 48 years. Find age of Anish.

(a) 36 (b) 20
(c) 40 (d) 12

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (a) Let the age of son = x Years
and age of Anish = 3x Years
$3x + x = 48$
$x = 12$
Therefore, age of aneesh = 3 × 12 = 36 Years

109. Five years ago, the son's age was one third of his mother's age at that time. If the sum of their present age is 70, find the present age of the mother.

(a) 45 (b) 55
(c) 50 (d) 60

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (c) Let the age of Mother before 5 years = x Years
$\therefore$ Present age of Mother = $(x + 5)$ Years
and age of son before 5 years $= \frac{x}{3}$ Years
$\therefore$ Present age of son $= \left(\frac{x}{3} + 5\right)$ Years
According to question,-
$$x + 5 + \frac{x}{3} + 5 = 70$$
$$\frac{4x}{3} + 10 = 70$$
$4x + 30 = 210$
$4x = 180$
$x = 45$
Present age of Mother = $x + 5$ Years
= 45 + 5 = 50 Years
Therefore, present age of Mother is 50 years.

110. Soham is 10 years younger than Parth. Before 8 years soham's age was three times of his present age that was 4 years more than twice of the age of Parth. Find Soham's present age.

(a) 28 years (b) 32 years
(c) 30 years (d) 33 years

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (b) : Let the present age of Soham = x years
Present age of Parth = (x + 10) years
Age of Soham before 8 years = (x – 8)
Age of parth before 8 years = x + 10 – 8 = (x + 2)
According to question,
$3(x - 8) = 2(x + 2) + 4$
$3x - 24 = 2x + 4 + 4$
x = 32 years
Present age of Soham = 32 years

111. Nilanjan is 11 years younger than Bharti. Fifteen years later, Bharti's age will be 1.2 times as that of Nilanjan's age. What is Nilanjan's present age?

(a) 38 Years (b) 40 Years
(c) 42 Years (d) 43 Years

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (b) : Let the present age of Bharti = x Years
Present age of Nilanjan = (x – 11) Years
According to question,-
$(x + 15) = (x - 11 + 15)\ 1.2$
$(x + 15) = (x + 4) \times 1.2$
$(x + 15) = 1.2x + 4.8$
$15 - 4.8 = 0.2x$
$10.2 = 0.2x$
x = 51 Years
Present age of Nilanjan = 51 – 11 = 40 Years

112. Srinivas is four times older than his daughter. Five years ago, Srinivas was nine times older than his daughter's age at that time. What is his daughter's present age?

(a) 8 Years (b) 6 Years
(c) 5 Years (d) 10 Years

RRB Group-D – 22/09/2018 (Shift-III)

Ans. (a) : Let the present age of daughter = x Years
Present age of Srinivas = 4x Years
According to question,
$9(x-5) = 4x-5$
$9x-45 = 4x-5$
$5x = 40$
$x = 8$
Therefore Present age of daughter = 8 Years

113. Eight years ago, Ashwin's age was 1 year less than 3 times of Arpit's age. Six years ago Ashwin's age was 1 year more than twice of Arpit's age. What will be the age of Arpit after 7 years?

(a) 19 Years (b) 15 Years
(c) 16 Years (d) 12 Years

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (a) : Let the age of Arpit before 8 years = x years
Age of ashwin = (3 x – 1) Years
Present age of Arpit = (x + 8) Years
and present age of Ashwin = (3x + 7) Years
According to question,-
$(3x+7-6) = 2(x+8-6)+1$
$3x+1 = 2x+4+1$
$x = 4$
Present age of arpit = x + 8 = 4 + 8 = 12 Years
Age of arpit after 7 years = 12+7 = 19 Years

114. After 19 years from now vinod's age will be twice of Anand's age. Seven years ago, Anand's age was 1/4 of Vinod's age. What is Vinod's present age?

(a) 53 Years (b) 57 Years
(c) 55 Years (d) 59 Years

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (d) Let the age of Anand before 7 years = x Years
Let the age of vinod 7 years ago = 4x Years
Present age of anand = (x + 7) Years
Present age of vinod = (4x + 7) Years
According to question,
$2(x+7+19) = (4x+7+19)$
$2x+52 = 4x+26$
$2x = 26$
$x = 13$
Present age of vinod = $4x+7 = 4\times 13+7 = 59$ Years

115. After 11 years, Raghav's age will be 5 times of his 5 years before age. What is the present age of Raghav?

(a) 7 Years (b) 8 Years
(c) 4 Years (d) 9 Years

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : Let, the present age of Raghav = x Years
Age of Raghav after 11 years = (x+11) Years
Age of Raghav before 5 years = (x-5) Years
According to question,
$(x+11) = (x-5)\times 5$
$x+11 = 5x-25$
$36 = 4x$
$x = 9$
∴ Present age of Raghav = 9 Years

116. Rani is two years older than Ragini, who is twice the age of Nag. If the age of Rani, Ragini and Nag is 27 years, then the age of Ragini will be?

(a) 9 (b) 8
(c) 10 (d) 7

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (c) Let the age of Ragini = x Years
Age of Rani = $(x+2)$ Years
Age of Naag = $\frac{x}{2}$ Years
According to question,-
$x+x+2+\frac{x}{2} = 27$
$2x+2x+4+x = 54$
$5x = 50$
$x = 10$ Years

117. Vishal is 20 years younger than Arya. After 12 years Arya's age will be 1.25 times of Vishal's age. Arya's age now is _____ years.

(a) 72 (b) 68
(c) 88 (d) 78

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (c) Let the age of Vishal = x Years
age of Arya = (x + 20) Years
According to question,
$(x+20+12) = (x+12)\ \frac{5}{4}$
$x+32 = \frac{5x+60}{4}$
$4x+128 = 5x+60$
$x = 68$
Therefore, Present age of Arya = x+20
= 20+68 = 88 Years

118. Five years before Rohit's age was 2/3 times of Rohan's age. After 5 years Rohan's age will be 5/4 times of Rohit's age. What is the present age of Rohit.

(a) 25 Years (b) 20 Years
(c) 10 Years (d) 15 Years

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (d) Let the Present age of Rohit = x Years
Present age of Rohan = y Years
According to first condition,
$(x-5) = (y-5)\times\frac{2}{3}$

$3x - 15 = 2y - 10$

$3x = 2y + 5$

$x = \frac{2y+5}{3}$ .....(i)

According to second condition–

$\frac{5}{4} \times (x+5) = (y+5)$

$5x + 25 = 4y + 20$

$5x - 4y = -5$

or, $5x = 4y - 5$

$x = \frac{4y-5}{5}$ .....(ii)

from equation (i) and (ii)

$\frac{2y+5}{3} = \frac{4y-5}{5}$

$10y + 25 = 12y - 15$

$2y = 40$

$y = 20$ Years

Putting the value of y in equation (i)

$x = \frac{2 \times 20 + 5}{3}$

$x = \frac{45}{3} = 15$ years

Therefore, Present age of Rohit = 15 Years

119. One year ago, Akash's father age was 9 times of Akash's age. After 3 years his father's age will be 5 times of his age. What will be the age of Akash next year?

(a) 6 Years (b) 8 Years
(c) 5 Years (d) 4 Years

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (a) Let the present age of Akash's father is x years and age of akash is y years.

According to question,

$(x-1) = 9(y-1)$

$x - 1 = 9y - 9$

$x - 9y = -8$ ..... (i)

Again after 3 years

$(x+3) = 5(y+3)$

$x + 3 = 5y + 15$

$x - 5y = 12$ .....(ii)

on solving both equation (i) and (ii)

$4y = 20$

$y = 5$

Therefore, the age of akash next years = 5 +1 = 6 Years

120. 5 years ago a man was 7 times older than his son. Now in 5 years, he will be 3 times older than his son. What was their age before two years?

(a) 26 Years and 10 Years
(b) 36 Years and 12 Years
(c) 38 Years and 8 Years
(d) 30 Years and 6 Years

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (c) Let the age of father before 5 years = 7x Years

Age of son before 5 years = x Years

Present age of father = (7x + 5) Years

Present age of son = (x + 5) Years

According to question,

$7x + 5 + 5 = 3(x + 5 + 5)$

$7x + 10 = 3x + 30$

$4x = 20$

$x = 5$

Present age of father = 7 × 5 + 5 = 40 Years

Present age of son = 5 + 5 = 10 Years

Age of father before 2 years from now = 40–2 = 38 Years

Age of son before 2 years from now = 10–2 = 8 Years

121. A person is 9 times older than his son. Two years after, the father will be one year less than 6 times of his son. Find their present age?

(a) 27 Years and 3 Years
(b) 30 Years and 6 Years
(c) 26 Years and 10 Years
(d) 36 Years and 12 Years

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (a) Let the present age of son = x Years

∴ Present age of father = 9x Years

from question–

$(9x + 2) = 6(x+2) - 1$

$9x + 2 = 6x + 12 - 1$

$9x - 6x = 12 - 1 - 2$

$3x = 9$

$\boxed{x = 3}$

Age of father = 9x = 9 × 3 = 27 Years

Age of son = x = 3 Years

122. Daliya is 20 years older than Neetu and in two years her age will be twice the age of Neetu. What is Neetu's present age.

(a) 18 Years (b) 16 Years
(c) 20 Years (d) 22 Years

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (a) Let the present age of Neetu = x Years

then Present age of Daliya = (20 + x) Years

According to question,

$[(20 + x) + 2] = 2(x+2)$

$22 + x = 2x + 4$

$\boxed{x = 18}$

Therefore, present age of Neetu = x = 18 Years

123. Before 5 Years Sindhu's age was three times as old Kaveri's age. 10 years later from now, Kaveri's age will be half of sindhu's age. After 5 years from now, what will be Kaveri's age.

(a) 15 (b) 20
(c) 55 (d) 25

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (d) : Let the present age of Kaveri = x Years
and Present age of Sindhu = y Years
According to question,
from first condition–
$y - 5 = 3(x - 5)$
$y - 5 = 3x - 15$
$3x - y = 15 - 5$
$3x - y = 10$ .....(i)
from second condition–
$\frac{(y+10)}{2} = (x+10)$
$y + 10 = 2x + 20$
$2x - y = 10 - 20$
$2x - y = -10$ .....(ii)
by subtracting equation (ii) from equation (i)
$3x - y = 10$
$2x - y = -10$
$- \quad + \quad +$
$3x - 2x = 20$
$x = 20$
Present age of Kaveri = 20 Years
Age of Kaveri after 5 years = 20 + 5 = 25 Years

124. The sum of Daniel and Dinara's age is 115 years. Five years ago two times of Dinara's age was equal to three times Daniel's age. What is the present age of Dinara?
(a) 62 Years (b) 64 Years
(c) 66 Years (d) 68 Years
RRB Group-D – 05/12/2018 (Shift-I)

Ans : (d) Let the present age of Deniel = x Years
and present age of Dinara = y Years
According to first condition–
$x + y = 115$ .....(1)
According to second condition–
$3(x-5) = (y-5) \times 2$
$3x - 15 = 2y - 10$
$3x - 2y = -10 + 15$
$3x - 2y = 5$.....(2)
By multiplying 3 in equation (i) and adding in equation (ii)-
$2x + 2y = 130$
$3x - 2y = 5$
$5x = 235$
$x = 47$
again from equation (i)–
$y = 115 - 47$
$y = 68$
Therefore, Present age of Dinara = y = 68 Years

125. The sum of father and mother's age is 7.5 times that of son's age. Mother's age is 35 years. If the father's age is 4 times the age of his son, then what is the son's age?
(a) 15 (b) 10
(c) 18 (d) 12
RRB Group-D – 04/12/2018 (Shift-III)

Ans. (b) According to question,
Father + Mother = 7.5 × age of son ...(i)
Age of mother = 35Years ...(ii)
Age of father = 4 × age of son ...(iii)
putting the value of equation (ii) and (iii) in equation (i)-
4×age of son + 35 = 7.5×age of son
(7.5 − 4) × age of son = 35
3.5 × age of son = 35
Age of son = 35/3.5 = 10 Years

126. The sum of the present age of two persons is seven times the difference of their age. After 5 years the sum of their age will be nine times the difference between their age. What is the present age of the elder person?
(a) 32 Years (b) 10 Years
(c) 20 Years (d) 35 Years
RRB Group-D – 15/11/2018 (Shift-II)

Ans : (c) Let the present age of both person is x and y years.
According to question,
$x + y = 7(x - y)$
$x + y = 7x - 7y$
$6x = 8y$
$\frac{x}{y} = \frac{8}{6}$
$\frac{x}{y} = \frac{4}{3}$ $x = 4k, y = 3k$
According to question,
$(x + 5) + (y + 5) = 9[x + 5 - (y + 5)]$
$x + y + 10 = 9(x + 5 - y - 5)$
$x + y + 10 = 9(x - y)$
$9x - x - 9y - y = 10$
$8x - 10y = 10$
Putting the value of x and y-
$\because x = 4k$
$y = 3k$
$32k - 30k = 10$
$2k = 10$
$k = 5$
$x = 4k$
$= 4 \times 5$
= 20 Years
$y = 3k$
$= 3 \times 5$
= 15 Years
Therefore, Age of eldest person = x = 20 Years

127. The product of the age of Anusha and Neelima is 240. If twice the age of Neelima is equal to the 4 more over the age of Anusha, so how old is Anusha?
(a) 18 Years (b) 16 Years
(c) 20 Years (d) 14 Years
RRB Group-D – 05/11/2018 (Shift-I)

Ans. (c) : Let the age of anusha = x Years
Age of Neelima = y Years
According to question,
$x \times y = 240$ ..... (i)
And, $x + 4 = 2y$
$y = \left(\frac{x+4}{2}\right)$ .....(ii)
putting the value of y in equation (i)
$x \times \left(\frac{x+4}{2}\right) = 240$
$x^2 + 4x = 480$
$x^2 + 4x - 480 = 0$
$x^2 + (24 - 20)x - 480 = 0$
$x^2 + 24x - 20x - 480 = 0$
$x(x + 24) - 20(x + 24) = 0$
$(x + 24)(x - 20) = 0$
$x = -24, 20$
Therefore, age of Anusha = x = 20 Years

128. 2/3of my age is 3/4 of my cousin's age. My age before 3 years was exactly what my cousin will be one year later. What is my present age (in years)?
(a) 36 (b) 18
(c) 45 (d) 27
RRB Group-D – 05/11/2018 (Shift-II)

Ans : (a) Let My present age is x years and age of cousin is y years-
According to question,
$\frac{2x}{3} = \frac{3y}{4}$
$8x - 9y = 0$ … (i)
And, $x - 3 = y + 1$
$x - y = 4$ … (ii)
by multiplying 9 in equation (ii) and subtracting from equation (i)–
$8x - 9y = 0$
$\underline{9x - 9y = 36}$
$-x = -36$
x = 36 Years

129. The present age of Raghu and Sita are 17 years and 41 years respectively. 5 years before Raghu's age was _____of Sita age.
(a) 3/4 (b) 1/3
(c) 1/2 (d) 2/3
RRB Group-D – 24/10/2018 (Shift-III)

Ans. (b) : Present age of Raghu = 17 Years
Present age of Sita = 41 Years
Age ratio of both before 5 years = $\frac{17-5}{41-5} = \frac{12}{36}$
= 1 : 3
Therefore, Raghu's age was 1/3 of Sita's age.

130. At present Bratin is 18 years old, while his cousin is 7 years old. In how many years Bratin's age will be 1.5 times that of his cousin's age?
(a) 13 (b) 14
(c) 16 (d) 15
RRB Group-D – 01/10/2018 (Shift-III)

Ans : (d) Let after x years, Bratin's age will be 1.5 times that of cousin's age.
(x+18) =1.5(x+7)
x+18 = 1.5x+10.5
0.5x = 7.5
x = 15

131. A man is 24 years older than his son. Four years later, his age will be twice of his son's age. What is the present age of father?
(a) 40 Years (b) 44 Years
(c) 42 Years (d) 48 Years
RRB NTPC 04.04.2016 Shift : 3

Ans : (b) Let the present age of father = x Years
∴ Present age of son = $(x - 24)$ Years
Age of father after 4 years = $(x + 4)$ Years
and age of son after 4 years = $(x - 20)$ Years
According to question,
$x + 4 = 2 \times (x - 20)$
$x + 4 = 2x - 40$
$2x - x = 4 + 40 \Rightarrow x = 44$
Therefore, Present age of father is 44 years.

132. At present, Ram is 4 times of his son's age. After 5 years he will be 3 times the age of his son. Find their present age?
(a) 60,15 (b) 40,10
(c) 20,5 (d) 32,8
RRB NTPC 28.03.2016 Shift : 1

Ans : (b) Let the Present age of Ram's Son = x Years
Present age of Ram = 4x Years
According to question,
$4x + 5 = 3(x + 5)$
$\Rightarrow 4x + 5 = 3x + 15 \Rightarrow x = 10$
Therefore, Present age of Ram and his son is 40 years and 10 years respectively.

133. After five years, Mayank age will be 3/5th of his father's age. Five years ago the ratio of their age was 2:5. Find the present age of Mayank.
(a) 17 (b) 13
(c) 19 (d) 15
RRB NTPC 19.04.2016 Shift : 3

Ans : (b) Let before 5 years Mayank's age is 2x years and father's age is 5x years.
Then after 5 years
$\frac{\text{Age of Mayank}}{\text{Age of Father}} = \frac{3}{5}$

$$\frac{2x+10}{5x+10}=\frac{3}{5}$$
$10x + 50 = 15x + 30$
$5x = 20$
$x = 4$
Therefore, Present age of Mayank $= 2x + 5$
$= 2 \times 4 + 5 = 13$ Years

134. 6 years ago a man's age was 5 times of his son's age. After 10 years, he will be 3 times of his son. What is his son's present age?
(a) 20 (b) 18
(c) 24 (d) 22
RRB NTPC 18.04.2016 Shift : 3

Ans : (d) Let the present age of father and son is x and y respectively.
from question-
$x-6 = 5(y-6) \Rightarrow x-5y = -24$ .....(i)
and
$x+10 = 3(y+10) \Rightarrow x-3y = 20$ ..... (ii)
from equation (i) and (ii)–
$x = 86, y = 22$
Therefore, Present age of son is 22 years.

135. 2 years after from today a man's age will be four times that of his son and 4 years after that the man's age will be 3 times of his son's age. After how many years will the father's age be twice of his son?
(a) 15 years (b) 16 years
(c) 17 years (d) 18 years
RRB NTPC 10.04.2016 Shift : 3

Ans : (d) Let the age of Man = x Years
and age of son = y Years
from question,
$(x + 2) = (y + 2) \times 4$
$x + 2 = 4y + 8$
$x = 4y + 6$ …….(i)
again
$(x+6) = (y + 6) \times 3$
$x + 6 = 3y + 18$
$x = 3y + 12$ ……..(ii)
from equation (i) and (ii)
$4y + 6 = 3y + 12$
$y = 6$ Years
Putting the value of y in equation (i)
$x = 4\times6+6$
$x = 30$ Years
Now, Let after A years, father's age will be 2 times that of his son.
$\therefore (30 + A) = (6 + A) \times 2$
$30 + A = 12 + 2A$
$A = 18$ years

136. The product of Sapna and Anubha's age is 150. If 4 times the age of Anubha is 10 years more than the age of Sapna. Find the age of Sapna.
(a) 20 (b) 27
(c) 19 (d) 17
RRB NTPC 22.04.2016 Shift : 1

Ans : (a) Let the age of Sapna = x Years
Age of anubha = y Years
According to question,
$xy = 150$ .....(i)
$4y = x + 10$ .....(ii)
from equation (i) $y = \frac{150}{x}$ putting in equation (ii)
$$4\times\frac{150}{x} = x + 10$$
$600 = x^2 + 10x$
$x^2 + 10x - 600 = 0$
$x^2 + 30x - 20x - 600 = 0$
$x(x+30) - 20(x+30) = 0$
$(x+30)(x-20) = 0$
$x + 30 = 0$
$x = -30$ (invalid)
$x - 20 = 0$
$x = 20$
Therefore, Age of Sapna = 20 Years

137. Bipul is 16 years younger than Saibal after. 12 years from now, Saibal's age will be 1.5 times that of Bipul. Saibal is now _____ years old.
(a) 42 (b) 45
(c) 40 (d) 36
RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (d) Let the present age of saibal = x years
then, Present age of Bipul = (x–16) years
Age of Saibal after 12 years = (x+12) Years
Age of Bipul after 12 years = (x–16+12) = (x–4) Years
According to question,
$(x+12) = 1.5 (x-4)$
$x+12 = 1.5x-6$
$0.5x = 18$
$x = \frac{18}{0.5}$
$x = 36$ Years
Therefore, Saibal's present age is 36 years.

138. 15 years ago, Shyam was twice as old as Prabhat. After 5 years from now Prabhat's age will be $\frac{5}{8}$ of Shyam's age. What is Shyam's present age?
(a) 72 years (b) 75 years
(c) 80 years (d) 64 years
RRB ALP & Tec. (31-08-18 Shift-III)

Ans : (b) Let the present age of prabhat = x years
Present age of shyam = y years
According to question,
$2(x-15) = (y-15)$
$2x-30 = y-15$
$2x-y = 15$ ..... (1)
Both age after 5 years from present,
$$(x+5) = \frac{5}{8}(y+5)$$

$8x+40 = 5y+25$

$8x-5y = -15$ ..... (2)

By multiplying 5 in equation (1)

$(2x-y)5 = 15\times5$

$10x-5y = 75$ ..... (3)

By subracting equation (3) from equation (2)−

$8x-5y = -15$

$10x-5y = 75$

$\underline{-\quad +\quad -}$

$-2x = -90$

$x = 45$

Putting the value of x in equation (1)–

$2 \times 45 - y = 15$

$y = 90-15$

$y = 75$ Years

There, Prsent age of Shyam is 75 years.

139. Jina is 24 years younger than her mother. Eight years after her mother's age will be $\frac{5}{3}$ times her age. What is Jina's present age (in years)?

(a) 24 (b) 22

(c) 26 (d) 28

RRB ALP & Tec. (30-08-18 Shift-III)

Ans : (d) Let the present age of Jeena = x Years

Present age of Mother = (x + 24) Years

Age of Jeena after 8 years = (x + 8) Years

Age of Mother after 8 years = (x+24+8) Years

According to question,

$x + 32 = \frac{5}{3}(x+8)$

$3x + 96 = 5x + 40$

$2x = 56$

$x = 28$

Therefore Present age of Jeena = 28 Years

140. Priyankur's present age is seven years less than thrice the age of his cousin Rihana. After sixteen years from now Priyakur's age will be 150% of that of Rihana. What is Priyankur's present age (in years)?

(a) 17 (b) 23

(c) 20 (d) 26

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (b) Let the Present age of Rihana = x Years

Then Present age of Priyankur = (3x–7) Years

According to question,

$(3x-7+16) = (x+16)\times\frac{150}{100}$

$(3x+9) = (x+16)\times\frac{3}{2}$

$6x+18 = 3x+48$

$6x-3x = 48-18$

$3x = 30$

$x = 10$

Present age of Priyankur = 3x–7 = 3×10–7

= 30–7= 23 Years

141. Jeremy is 26 years younger than his father. Eight years after his father's age will be two years less than twice his age. What is Jeremy's present age (in years)?

(a) 20 (b) 24

(c) 22 (d) 18

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (a) Let the present age of Jeremyi = x Years

Present age of Father = (x + 26) Years

According to question,

Then, $2(x + 8) - 2 = (x + 26 + 8)$

$2x + 16 - 2 = x + 34$

$x = 34 - 14$

$x = 20$ Years

142. The sum of the present age of a father and his son is 60 years. Six years ago, father's age was five times the age of the son. After 6 years, son's age will be:

(a) 20 years (b) 21 years

(c) 15 years (d) 19 years

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (a) Let the age of son = x Years

Then age of father = (60 – x) Years

Age before 6 years

According to question,

$5(x - 6) = (60 - x - 6)$

$5x - 30 = 54 - x$

$6x = 84,$

$x = 14$

Age of son after 6 years = 14 + 6 = 20 Years

143. Satheesh is two years older than Gautam who is twice as old as Sai. If the total of the ages of Satheesh, Gautam and Sai is 27, then how old is Gautam?

(a) 12 (b) 10

(c) 11 (d) 13

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (b) Let the age of Sai = x Years

Age of Gautam = 2x Years

And age of Satish = (2x+2) Years

According to question,

$x+2x+2x+2 = 27$

$5x= 27-2$

$x = \frac{25}{5} = 5$

So, Age of Gautam = 2x = 2×5= 10 Years

144. John is 15 years younger than Jill. 12 years ago, Jill's age was 1.5 times that of John. Jill is now .....years old.

(a) 57 (b) 45

(c) 30 (d) 42

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (a) Let the age of John is x years and age of Jill is y years.

According to question,

$y - x = 15$ .....(i)

and $(y-12) = (x-12)\,1.5$

$y - 12 = 1.5x - 18$

$y - 1.5x = -6$ .....(ii)

From equation (i) and (ii)–

$y - x - (y - 1.5x) = 15 - (-6)$

$y - x - y + 1.5x = 15 + 6$

$0.5x = 21$

$x = \frac{21}{0.5}$

$x = 42$

putting the value of x in equation (i)–

$y - 42 = 15$

$y = 15+42 = 57$

So, Present age of Jill is 57 years.

145. Roshan's present age is three years less than 1.5 times that of Usha's. 12 years ago Usha's age was three years more than half of Roshan's age. What is Roshan's present age in years?

(a) 42 (b) 39

(c) 33 (d) 30

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (a) Let the present age of Roshan is x years and present age of Usha is y years.

According to first Condition–

$x = 1.5y - 3$ ..... (i)

According to second Condition–

$\frac{x-12}{2} + 3 = y - 12$ ..... (ii)

Putting the value of x in equation (ii) from equation (i),

$\frac{(1.5y-3)-12}{2} + 3 = y - 12$

$1.5y - 15 + 6 = 2y - 24$

$0.5y = 15$

$y = 30$

Putting y = 30 in equation (i)

$x = 1.5 \times 30 - 3$

$x = 45.0 - 3$

$x = 42$ Years

So present age of Roshan = 42 years.

146. 13 years ago Ram was twice as old as Sunny. Three years from now Sunny's age will be $\frac{3}{5}$ of Ram's age. What is Ram's present age?

(a) 64 years (b) 72 years

(c) 80 years (d) 77 years

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (d) If present age of Ram = x Years

and Present age of sunny = y Years

Age of Ram and Sunny before 13 years would be (x-13) and (y-13) respectively–

from question,

$(x-13) = (y-13) \times 2$

$x - 13 = 2y - 26$

$x - 2y = -13$ .....(i)

Age of Ram and Sunny after 3 years will be (x+3) and (y+3) respectively.

$(x+3) \times \frac{3}{5} = y+3 \Rightarrow 3x + 9 = 5y + 15$

or $3x - 5y = 6$ .....(ii)

By Multiplying 3 in equation (i) and subtracting from equation (ii)–

$3x - 6y = -39$

$3x - 5y = 6$

$y = 45$ Years

Putting the value of y in equation (i),

$x - 2 \times 45 = -13$

$x - 90 = -13$

$x = 77$ years

147. Three years from now Dharitri's age will be eight years less than twice of Yunees age. The sum of their present ages is 61 years. What is Dharitri's present age?

(a) 43 years (b) 36 years

(c) 41 years (d) 39 years

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (d) Let the present age of Dharitri and Yunees is x, y years respectively.

According to question,

$x + 3 = 2(y + 3) - 8$

$x + 3 = 2y + 6 - 8$

$x + 3 = 2y - 2$

$x - 2y = -5$ .....(i)

and $x + y = 61$ .....(ii)

By subtracting equation (ii) from equation (i),

$x - 2y = -5$

$x + y = 61$

$-3y = -66$

$y = 22$

Putting y = 22 in equation (ii),

$x + 22 = 61$

$x = 39$

So, Present age of Dharitri is 39 years.

148. Seven years from now Virat will be twice as old as Mohinder. Five years ago Mohinder's age was one years less than $\frac{2}{5}$ of Virat's age. What is Virat's present age?

(a) 53 years (b) 51 years

(c) 57 years (d) 55 years

RRB ALP & Tec. (09-08-18 Shift-I)

Ans : (d) Let the age of Virat before 5 years = x years

Age of Mohindar before 5 years $= \left(\frac{2x}{5} - 1\right)$ Years

Present age of Virat = (x + 5) Years

Present age of Mohindar $= \left(\frac{2x}{5} + 4\right)$ Years

According to question,

$x + 5 + 7 = 2\left(\frac{2x}{5} + 4 + 7\right)$

$5(x + 12) = 4x + 110$, $x = 50$

Present age of Virat = x + 5 = 50 + 5 = 55 Years

Type - 7

149. The average age of P and Q is 24 years. Average age of P, Q and R is 22 years. Find the sum of their age in the previous year?

(a) Data is insufficient (b) 90
(c) 87 (d) 95

RRB JE - 27/05/2019 (Shift-III)

Ans : (a) P + Q = 24 × 2 ..... (i)
P + Q + R = 22 × 3 ..... (ii)
By subtracting equation (i) from equation (ii)–
R = 66 – 48 = 18 Years
The different ages of P and Q are not knwon so the data is insufficient to answer the question.

150. The sum of the ages of father and son is 50. 6 years before the age of father was 6 more than thrice of the age of son. What will be the fathers age after 6 years.

(a) 40 Years (b) 48 Years
(c) 42 Years (d) 50 Years

RRB JE - 28/05/2019 (Shift-I)

Ans : (c) Let the age of father = x Years
and age of son = y Years
x + y = 50 ---- (i)
According to question,
(x–6) = (y –6) ×3 + 6
x – 6 = 3y – 18 + 6
x – 3y = –6 ----(ii)
From equation (i) and (ii)-
x + y = 50
x – 3y = –6
– + +
4y = 56
y = 14
Putting the value of y in equation (i),
x = 36
So, Age of father after 6 years = 36 + 6 = 42 years.

151. The age of father is 5 years more than the mother age. Mother's present age is thrice of the daughter's age. The Present age of daughter is 12 years. What was the age of father at the time of daughter's birth?

(a) 29 Years (b) 25 Years
(c) 31 Years (d) 32 Years

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (a) ∵ Present age of daughter = 12 Years
So Present age of Mother = 3 × 12 = 36 Years
Given, present age of father = 36 + 5 = 41 Years
Therefore, at the birth of daughter the father's age
= 41–12 = 29 Years

152. A father says to his son "that at the time of your birth my age was as your present age. If the father's present age is 40 years, then what was the son's age before 5 years?

(a) 15 Years (b) 13 Years
(c) 17 Years (d) 23 Years

RRB Group-D – 31/10/2018 (Shift-II)

Ans. (a) Let the age of father at the birth of son = x years
So present age of son = x Years
Given, present age of father = 40 Years
$\Rightarrow (40 - x) = x$
$40 = 2x$
$x = 20$ Years
∴ Age of son before 5 years = 20 - 5 = 15 Years
So, before 5 years son's age was 15 years.

153. At the time of marriage, the age of a person was 6 year more than the age of his wife but after 12 years of marriage His age is 1.2 times of his wife's age. What was their age during the marriage?

(a) 27 Years, 18 Years
(b) 24 Years, 18 Years
(c) 21 Years, 18 Years
(d) 23 Years, 19 Years

RRB Group-D – 22/10/2018 (Shift-II)

Ans : (b) Let the age of the Man's wife at the time of marriage = x Years
and Age of Man = (x + 6) Years
∴ Age of man after 12 years = x+6+12 = (x+18) Years
According to question, x + 18 = (x + 12) × 1.2
x + 18 = 1.2x + 14.4
0.2x = 3.6
x = 18
So the age of the Man's wife = x = 18 Years
And age of Man = (x + 6) = (18 + 6) = 24 Years

154. The present age of Z is half of A's age. After 5 years the ratio of A and Z will be 11:6. After 3 years what will be the age of Z?

(a) 25 (b) 30
(c) 28 (d) 22

RRB NTPC 31.03.2016 Shift : 1

Ans : (c) If the present age of Z and A is x and 2x years.
According to question, $\frac{2x+5}{x+5} = \frac{11}{6}$
12x + 30 = 11x + 55
x = 25
Age of Z after 3 years = x + 3 = 28 Years

155. Rathin is now 16 years old while his cousin is 7 years. After how many years will Rathin's age be 1.5 times that of his cousin?

(a) 12 (b) 11
(c) 9 (d) 10

RRB ALP & Tec. (13-08-18 Shift-I)

Ans. (b) Let after x years, Rathin's age will be 1.5 times that of his Cousin's age.
Age of Rathin After x years = (x + 16) Years
Age of Cousin = (x+7) Years
(x+16) = (x+7)×1.5
x+16=1.5x+10.5
5.5=0.5x
$x = \frac{5.5}{0.5}$
x = 11 years

Type - 7

156. At the time of birth of Priya her father's age was 38 years. At the time of birth of her 4 year younger brother her mother's age was 36 years. How old is his mother younger than his father?

(a) 4 (b) 5
(c) 8 (d) 6

RRB JE - 23/05/2019 (Shift-II)

Ans : (d) According to question,
Age of father = 38 Years
∵ Mother's age during birth of 4 years younger brother = 36 Years
∴ Mother's age at priya's birth = 36 – 4 = 32 Years
∴ Age of father – age of mother = 38 – 32 = 6 Years

157. The average age of three children born at two years interval is 8 years. What is the age of the eldest child?

(a) 12 Years (b) 7 Years
(c) 10 Years (d) 8 Years

RRB JE - 24/05/2019 (Shift-II)

Ans : (c) Let the three children are A, B and C
A = x, B = x + 2, C = x + 4
According to question,

$$\frac{x+x+2+x+4}{3}=8$$

x + x + 2 + x + 4 = 24
3x = 18
x = 6
So, the age of eldest children = x + 4 = 6 + 4 = 10 Years

158. The average sum of 5 children born at 4 years interval is 80. What is the age of the eldest child?

(a) 18 (b) 24
(c) 16 (d) 28

RRB JE - 26/05/2019 (Shift-I)

Ans : (b) Let the five children are A, B, C, D and E
∵ A = x year
B = x + 4
C = x + 8
D = x + 12
E = x + 16
According to question,
x+ x+ 4+x+8+x+12+x+16 = 80
5x + 40 = 80
5x = 40
x = 8
∴ Age of eldest children = x + 16 = 8 + 16
= 24 Years

159. In a group of students, age of 1/5 students is less than 8 years. 2/5 students of the rest students is more than 8 years. What part of the groups age is exact 8 years?

(a) 4/25 (b) 12/25
(c) 2/5 (d) 3/5

RRB JE - 26/05/2019 (Shift-II)

Ans. (b)
Let the total number of students in the group = x
Number of students under 8 yers of age = $\frac{x}{5}$
Number of students over 8 years of age

$$=\left(x-\frac{x}{5}\right)\times\frac{2}{5}=\frac{8}{25}x$$

Number of students of 8 years =

$$x-\left(\frac{x}{5}+\frac{8x}{25}\right)=x-\frac{13x}{25}=\frac{12x}{25}$$

Part of exactly 8 years old students in the group = $\frac{12}{25}$

160. I have a brother who is 3 years elder to me when my brother was born then my sister was 6 years old. Our average age is 14. Now what is the age of my sister?

(a) 20 Years (b) 19 Years
(c) 17 Years (d) 18 Years

RRB RPF Constable – 19/01/2019 (Shift-III)

Ans : (b) Let the present age of Man = x Years
Present age of Man's brother = (x + 3) Years
Present age of Man's Sister · (x + 3 + 6) = (x + 9) Years
∴ According to question,
x + x + 3 + x + 9 = 3 × 14
3x + 12 = 3 × 14
x + 4 = 14
x = 10 Years
∴ Age of sister = 10 + 9 = 19 Years

161. The age ratio of two brothers is 7:15. The LCM of their age is 210. What is the age of elder brother?

(a) 45 (b) 42
(c) 15 (d) 30

RRB JE - 30/05/2019 (Shift-III)

Ans : (d) Let the age of younger brother is 7x years and age of elder brother is 15x years.
According to question,
L.C.M. of Number 7x and 15x = 105x
105 x = 210
x = 2
Age of elder borther = 15x = 15 × 2 = 30 Years

162. If 4 is subtracted from any person's age and is divided by 5, then the result represents the age of his grand son. There is a sister of grand son whose age is 5 years, 6 years less than the grandson. What is the age of grand father?

(a) 58 (b) 57
(c) 60 (d) 59

RRB JE - 31/05/2019 (Shift-III)

Ans. (d) Let the age of grandfather = x Years
According to question,
Age of Daughter = 5 Years
∴ Age of son = 11 Years
$\frac{x-4}{5}=11$
x – 4 = 55
x = 59 Years

163. A grand father and his grand daughter's age ratio is 9:2. The sum of their age is a whole number. The difference of their age may be multiple of 11, then find their age?
(a) 90, 20 (b) 95, 25
(c) 72, 16 (d) 99, 22
RRB JE - 02/06/2019 (Shift-II)

Ans. (d) According to question,
Age of grandfather = 9x Years
Age of grand daughter = 2x Years
$9x + 2x = x^2$
$11x = x^2$
x = 11
Age of grand father = 9x = 9 × 11 = 99 Years
Age of grand daughter = 2x = 2×11 = 22 Years

164. The age of a father is twice of his son's age. The HCF of their age is 22. What is the age of the son?
(a) 18 Years (b) 24 Years
(c) 22 Years (d) 20 Years
RRB JE - 02/06/2019 (Shift-I)

Ans : (c) Let age of son = x Years
then, age of father = 2x Years
HCF of x and 2x = x = x × 1
2x = x × 2 × 1
So, HCF = x
Therefore age of son = x Years = 22 Years

165. The total age of A and B is 12 years more than the total age of B and C. Who is the younger in A and C and By how many years?
(a) A, 6 Years (b) A, 12 Years
(c) C, 12 Years (d) C, 6 Years
RRB JE - 28/06/2019 (Shift-III)

Ans. (c) According to question,-
A + B = B + C + 12
A = C + 12
Therefore, It is clear that C is 12 years younger than A.

166. A grand father's age is 5 times more than the age of grandson. Which of the following numbers not support the approximate of the total age of the grand father and his grandson?
(a) 50 (b) 54
(c) 72 (d) 66
RRB RPF SI – 10/01/2019 (Shift-III)

Ans : (a) Let the age of Grandson = x Years
Age of Grandfather = 5x Years
Total age = 6x Years
Since Divisible Numbers from 6 are 54, 72 and 66.
Therefore, option (a) does not support of total age.

167. The age of P and Q is 50 and 40 respectively. Before how many years the ratio of their age was 3:2?
(a) 20 Years (b) 10 Years
(c) 15 Years (d) 5 Years
RRB JE - 02/06/2019 (Shift-I)

Ans : (a) Let Ratio of their age before x years was 3 : 2
According to question,-
$\frac{50-x}{40-x}=\frac{3}{2}$
100 – 2x = 120 – 3x
x = 20 Years

168. Sum of the age of two sisters is 81. Their age ratio is 4:5. How old is one sister than other?
(a) 9 (b) 27
(c) 18 (d) 12
RRB JE - 27/06/2019 (Shift-I)

Ans : (a) Let the age of both sisters is 4x, 5x
According to question,
4x + 5x = 81
9x = 81
x = 9
Age of first sister = 4x = 4 × 9 = 36 Years
Age of second sister = 5x = 5 × 9 = 45 Years
So, first sister is 9 years younger than second.

169. The age of Sita is twice the average age of Ram, Mohan and Sita. Ram's age is half of the average age of Ram, Mohan and Sita. If Mohan's age is 5 years then what is the average age of Ram, Mohan and Sita?
(a) 10 Years (b) 8 Years
(c) 7 Years (d) 15 Years
RRB Group D 07/12/2018 (Shift-I)

Ans : (a) Age of sita = $\frac{\text{Ram}+\text{Mohan}+\text{Sita}}{3}\times 2$
Given Mohan = 5 Years
3 × Sita = (Ram + 5 + Sita) × 2
3 × Sita = 2 × Ram + 10 + 2 × Sita
Sita = 2 × Ram + 10 -------(1)
Age of Ram = $\frac{\text{Ram}+\text{Mohan}+\text{Sita}}{3}\times\frac{1}{2}$
6 × Ram = Ram + 5 + Sita
5 × Ram = 5 + Sita
Sita = 5 × Ram – 5 -------(2)
By subtracting equation (1) from equation (2)
Age of Ram = 5, Aage of Sita = 20, Age of Mohan = 5
So Average Age = $\frac{\text{Ram}+\text{Sita}+\text{Mohan}}{3}$
$=\frac{5+20+5}{3}=10$ years

170. If 2/3 Children are in the age group of 1-12, 1/2 children are in the age group of 1-8. Then find the part of children in the age group of 9-12?
(a) 1/3 (b) 1/4
(c) 1/6 (d) 1/2
RRB NTPC 28.03.2016 Shift : 1

Ans : (c) The part of Students of 9-12 age group = part of student of (1-12) years – part of student of (1-8) years.

$$= \frac{2}{3} - \frac{1}{2} = \frac{4-3}{6} = \frac{1}{6}$$

171. The sum of the ages of 4 children born at the intervals of 4 years is 48. Find the age of the youngest child.

(a) 4 years (b) 5 years
(c) 6 years (d) 7 years

RRB NTPC 05.04.2016 Shift : 3

Ans : (c) All four children are x_1, x_2, x_3 and x_4.
Let the age of youngest $(x_1) = x$ Years
The age of older than that boy $(x_2) = (x + 4)$ Years
The age of older than that of that boy
$(x_3) = (x + 8)$ Years
The age of eldest boy $(x_4) = (x + 12)$ Years
$\therefore x + x + 4 + x + 8 + x + 12 = 48$ Years
$4x + 24 = 48$ Years
$4x = 24 \Rightarrow x = 6$ Years
$\therefore$ Age of the youngest boy = 6 Years

172. The average age of mother, father and two children is 30 years and 8 years respectively then find the average age of the family?

(a) 16 Years (b) 19 Years
(c) 18 Years (d) 17 Years

RRB NTPC 05.04.2016 Shift : 3

Ans : (b) Average age of family $= \frac{30\times2+8\times2}{4}$

$$= \frac{60+16}{4} = \frac{76}{4} = 19 \text{ Years}$$

173. 3 years before the age of John was 12 years and the age of Shankar was 15 years. What will be their average age after 5 years from today?

(a) 21.5 (b) 22.5
(c) 18.5 (d) 19.5

RRB NTPC 31.03.2016 Shift : 3

Ans : (a)
Age of John after 5 years from present
= 12 + 3 + 5 = 20 years
Age of Shankar after 5 years from present
= 15 + 3 + 5 = 23 years
Intended Average = $\frac{20+23}{2} = \frac{43}{2} = 21.5$ Years

174. The average age of 27 students in a class is 22. If the teacher's age is also included then average is raised by one. Then find the age of the teacher?

(a) 42 (b) 48
(c) 50 (d) 52

RRB NTPC 29.03.2016 Shift : 1

Ans : (c)
Total sum of age of 27 student
= 27 × 22 = 594 years
Total sum of age of (teacher + students)
= 28 × 23 = 644 Years
$\therefore$ Age of teacher = 644 – 594 = 50 Years

175. The average age of a group of 19 members is 24 years. If the age of the faculty is also included then the average is raised by 4 months. What is the age of faculty?

(a) 35 Years
(b) 30 Years 8 months
(c) 37 Years 4 months
(d) 32 Years 8 months

RRB NTPC 19.04.2016 Shift : 2

Ans : (b) Total age of 19 members of a group = 19×24 = 456 Years

Total age of 19 members with faculty $= 20\times\left(24+\frac{4}{12}\right)$

$$= 20\times\frac{73}{3}$$

$\therefore$ Age of faculty $= 20\times\frac{73}{3} - 456$

$$= \frac{1460-1368}{3} = \frac{92}{3} = 30\frac{2}{3} \text{ Years}$$

$= 30\text{Years} + \frac{2}{3}\times12$ months = 30 Years 8 months

176. The product of the age of swati and Aparna is 120. If thrice the age of Aparna is 2 years more than swati's age, then find the age of swati?

(a) 18 (b) 20
(c) 24 (d) 16

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) Let the age of swati is x years and age of Aparna is y years.
According to first condition–
$xy = 120$ ……..(i)
According to second condition–
$3y = x + 2$
$y = \left(\frac{x+2}{3}\right)$
Putting the value of y in equation (i)–
$x \times \left(\frac{x+2}{3}\right) = 120$
$x^2 + 2x - 360 = 0$
$x^2 + 20x - 18x - 360 = 0$
$x(x + 20) - 18(x + 20) = 0$
$(x + 20)(x - 18) = 0$
$x + 20 = 0$ or $x - 18 = 0$
$x = -20$ (Invalid) or $x = 18$
Therefore age of Swati = 18 Years

177. If in class 1st the average age of 20 students is 10 years and in class IInd the average age of 25 students is 12 years. What is the average age of all the students (in years)?

(a) 11 (b) 11.111
(c) 10.50 (d) 10.85

RRB NTPC 27.04.2016 Shift : 3

Ans : (b) Average age of all students

$$= \frac{\text{Total sum of age of all students}}{\text{Number of all students}}$$

$$= \frac{20\times10+25\times12}{20+25} = \frac{200+300}{45}$$

$$= \frac{500}{45} = 11.111$$

178. 15 years ago Cynthia was thrice as old as Brittany. The sum of their present ages is 94 years. How old is Brittany now?

(a) 33 Years (b) 32 Years
(c) 30 Years (d) 31 Years

RRB ALP & Tec. (30-08-18 Shift-I)

Ans : (d) If present age of Cynthia is x years and present age of Brittany is y years. Then,

x + y = 94 Years .....(i)

Their age before 15 years was (x-15) years and (y-15) years respectively. Then,

According to question

(x – 15) = 3(y – 15)

x – 15 = 3y – 45

x – 3y = –30 .....(ii)

By multiplying 3 in equation (i) and adding equation (ii)–

3x + 3y = 282

x – 3y = –30

4x = 252

x = 63 Years

Present age of Brittany ⇒ 63 + y = 94

or y = 94 – 63 = 31 years

179. The present age of Kavita, Rajita and Harita are in the ratio of 4 : 7: 9. Eight years ago, the sum of their age was 56. Find their present age (in years) :

(a) 16, 36, 28 (b) 16, 28, 36
(c) 20, 35, 45 (d) 12, 21, 27

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (b) Let the age of Kavita, Rajita and harita is 4x, 7x, 9x years respectively–

4x, 7x, 9x Years

According to question,

$$(4x-8)+(7x-8)+(9x-8)=56$$

$$20x-24=56$$

$$x=4$$

So present age of kavita = 4x= 4×4 =16 Years

Present age of Rajita = 7x = 7×4 = 28 Years

Present age of Harita = 9x = 9×4 = 36 Years

180. What is the age of the eldest sister if, the sum of age of 5 sisters born at the intervals of 3 years each is 50 years?

(a) 16 years (b) 4 years
(c) 14 years (d) 18 years

RRB ALP & Tec. (13-08-18 Shift-III)

Ans. (a) Let the age of eldest sister is x years then, age of other 4 sisters born at 3 years gap is (x-3), (x-6), (x-9) and (x-12) years respectively–

According to question,

x + x - 3 + x - 6 + x - 9 + x - 12 = 50

5x = 50 +30

5x = 80

x = 16 Years

181. The sum of one third of Poojith's age of three years ago and half of her present age after two years is 20 year. What is her present age?

(a) 23 years (b) 24 years
(c) 26 years (d) 25 years

RRB ALP & Tec. (10-08-18 Shift-III)

Ans : (b) Let the present age of Pujeetha = x Years

Age of Pujeetha before 3 years = (x – 3) Years

Age of Pujeetha after 2 years = (x + 2) Years

According to question,

$$\left(\frac{x-3}{3}\right)+\left(\frac{x+2}{2}\right)=20$$

$$\frac{2x-6+3x+6}{6}=20$$

$$\frac{5x}{6}=20$$

x = 6 × 4

x = 24 Years

Present age of Pujeetha = 24 Years

182. 3/5th of my present age is same as 5/6th of one of my cousin's age. My age before ten years will be his four years later age from now. What is my present age?

(a) 55 (b) 45
(c) 60 (d) 50

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Ans : (d) Let my present age = x Years

Age of cousin = y Years

According to question,

$$\frac{3x}{5}=\frac{5y}{6}$$

18x = 25y

18x – 25y = 0 ---------- (I)

And, x–10 = y + 4

x – y = 14 ----------(II)

From equation (II) × 25 – equation (I)

25x – 25y = 350

18x – 25y = 0

– + –

7x = 350

x = 50 Years

Therefore, my present age is 50 years.